

From black box to grey box modelling:

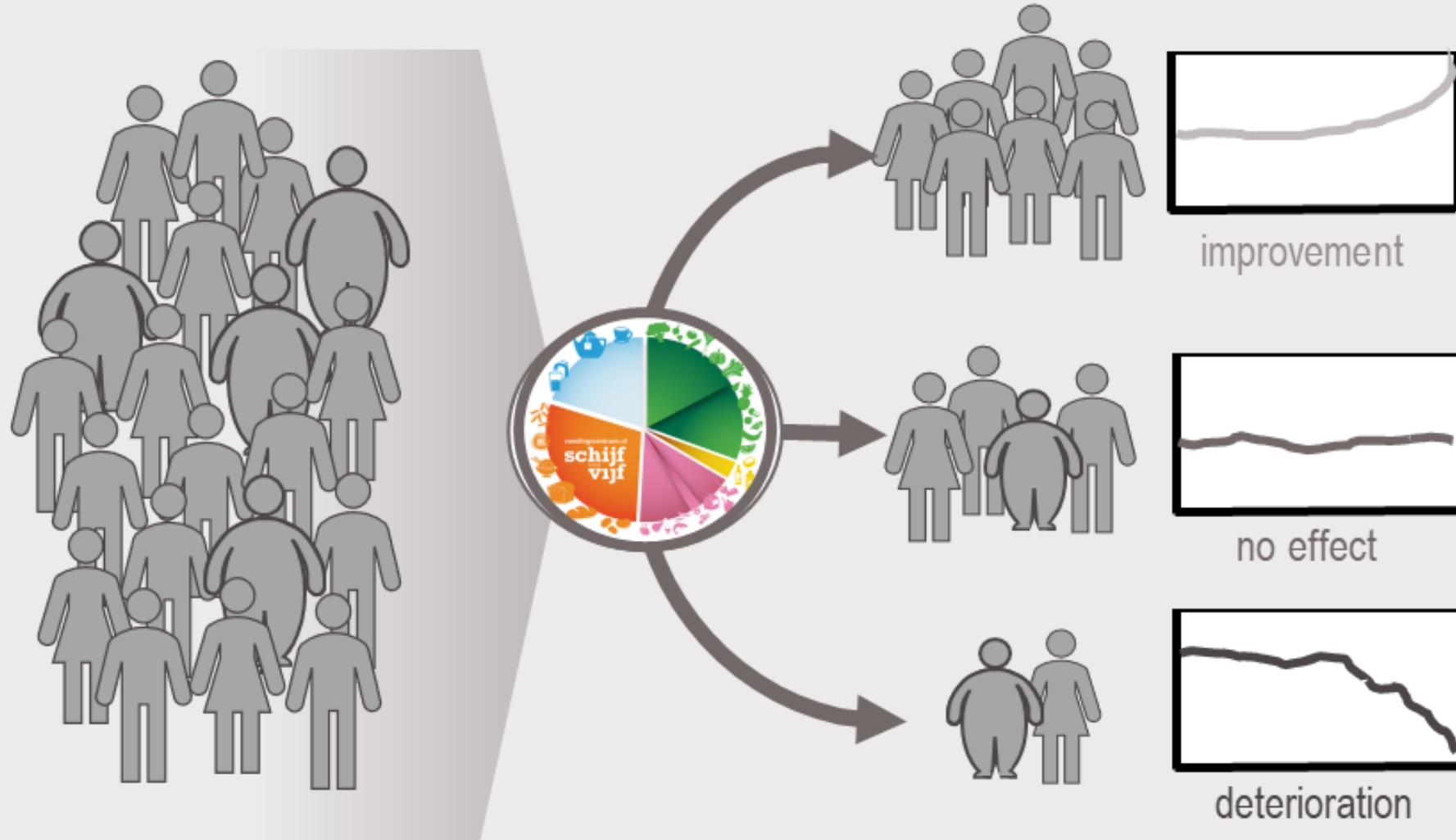
Hybrid modelling approaches for personalised medicine.

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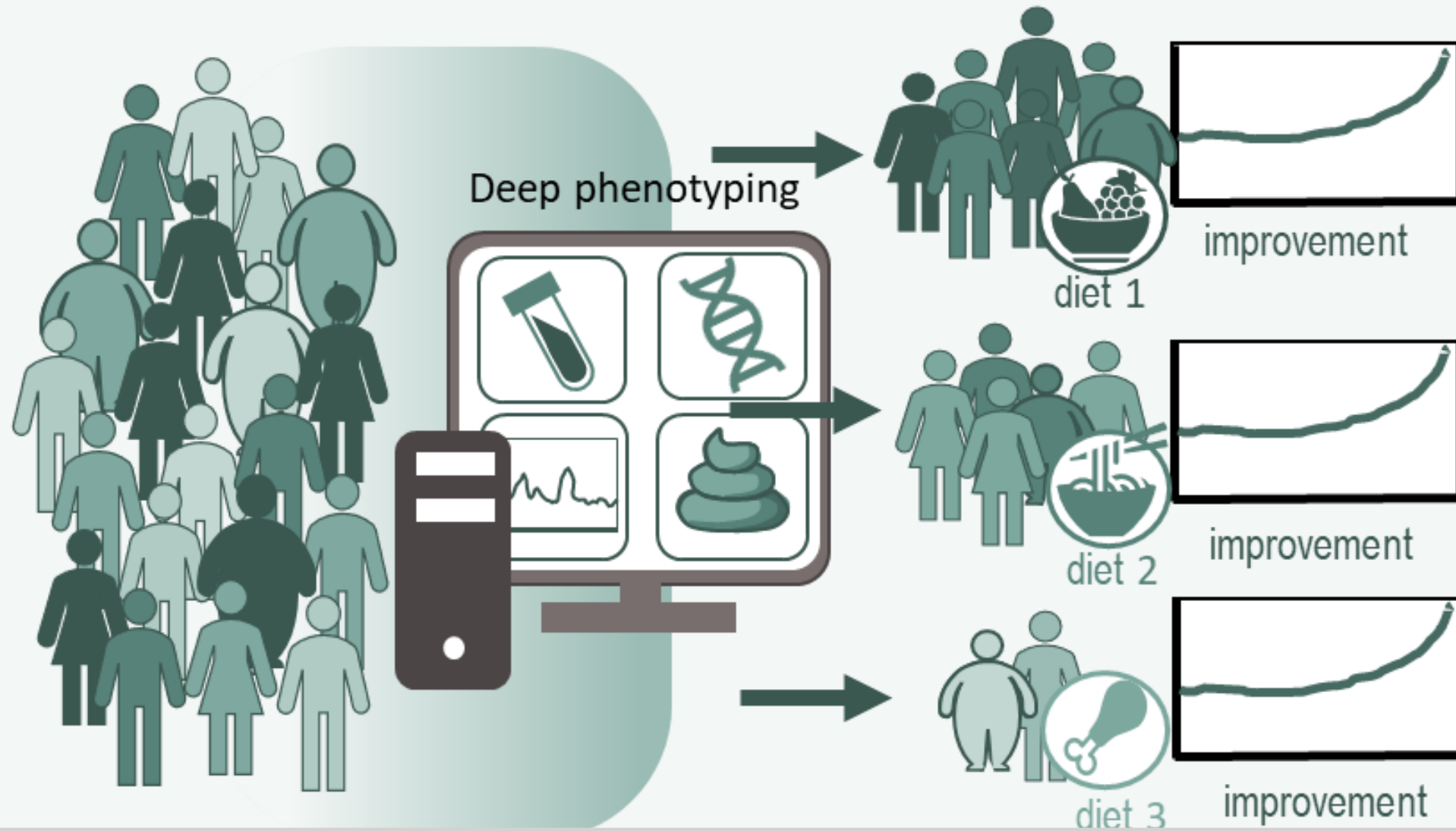
Current medicine

One-size-fits-all



Future of medicine

Personalised treatment strategies



BIGGER challenges

- High dimensional data.
- Sparse and noisy data.
- Small sample sizes.



WHAT IS HEALTH?

Health was once considered the **absence of disease or injury**.

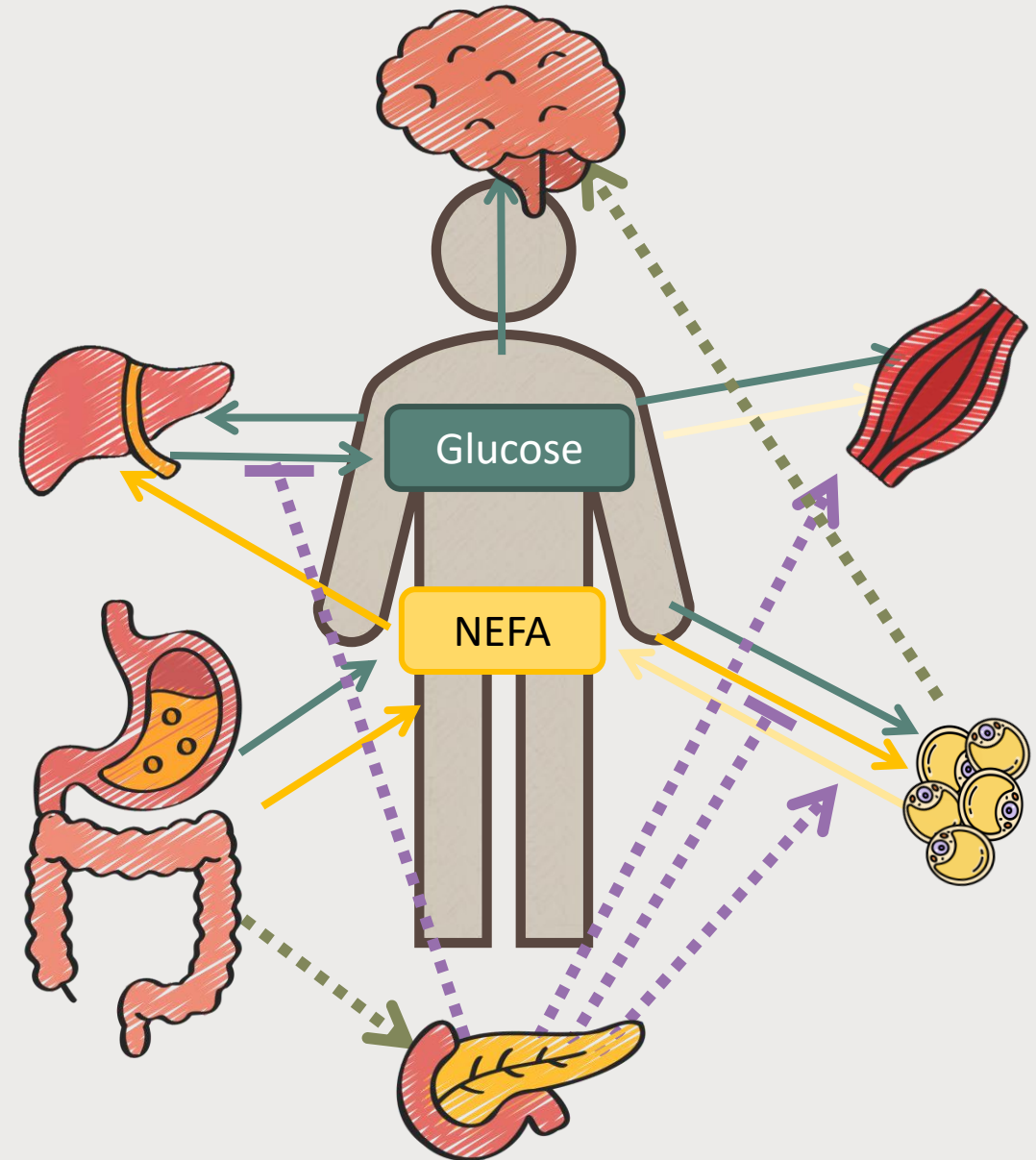
More recently, the concept of health has been redefined as the **ability** of an individual to **respond and adapt** to **physical, emotional, or social challenges**, referred to as **resilience**.



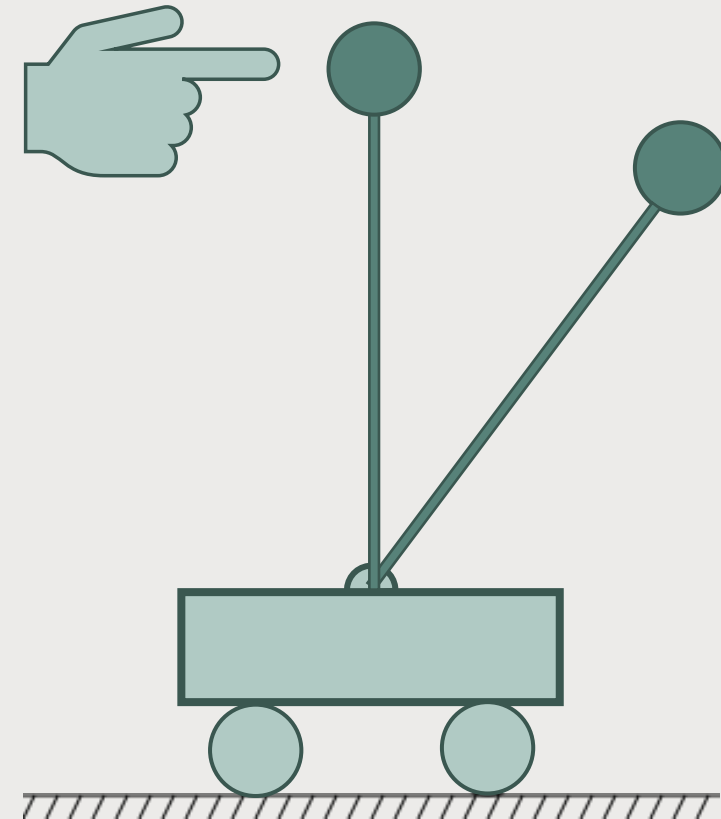
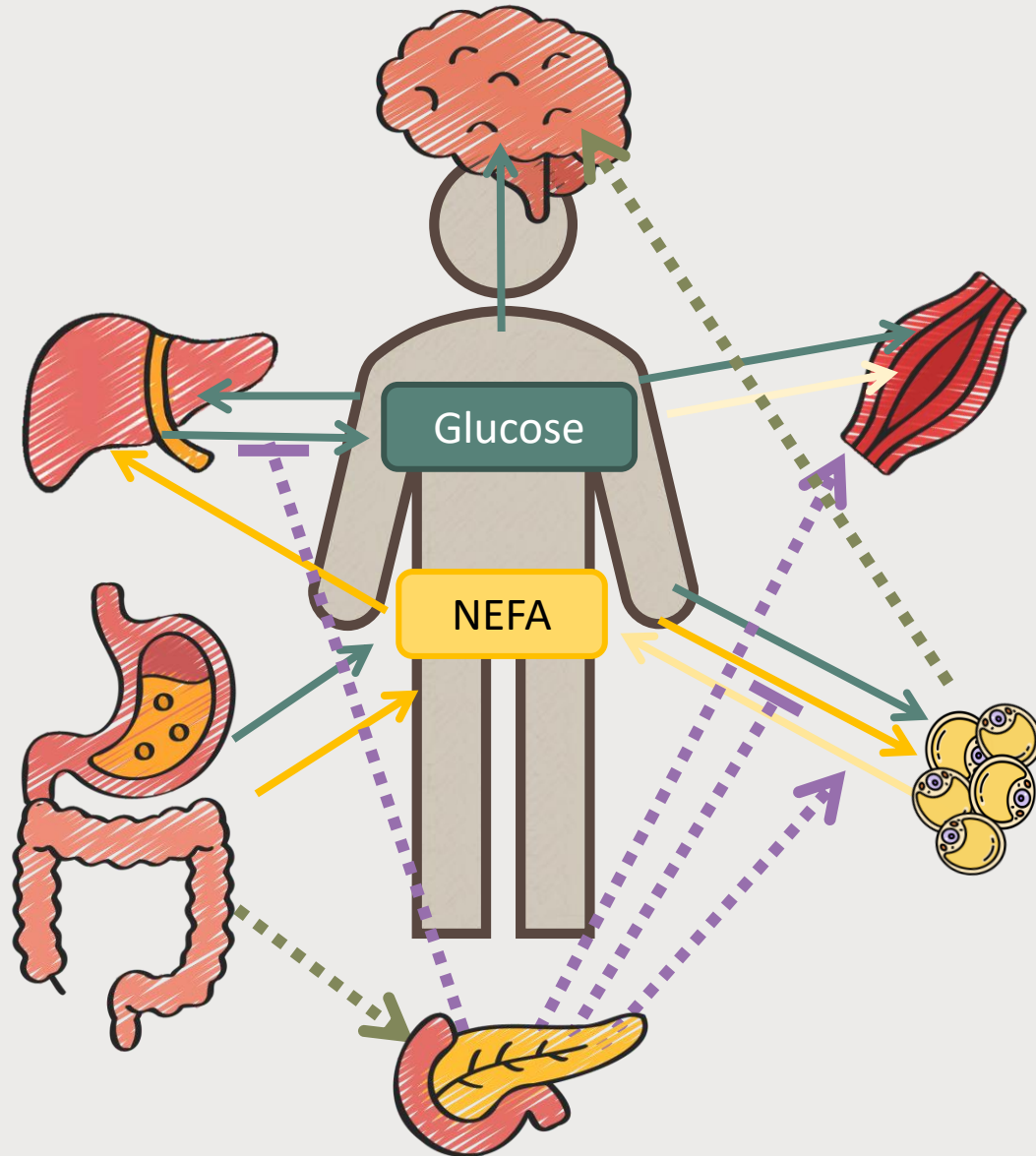
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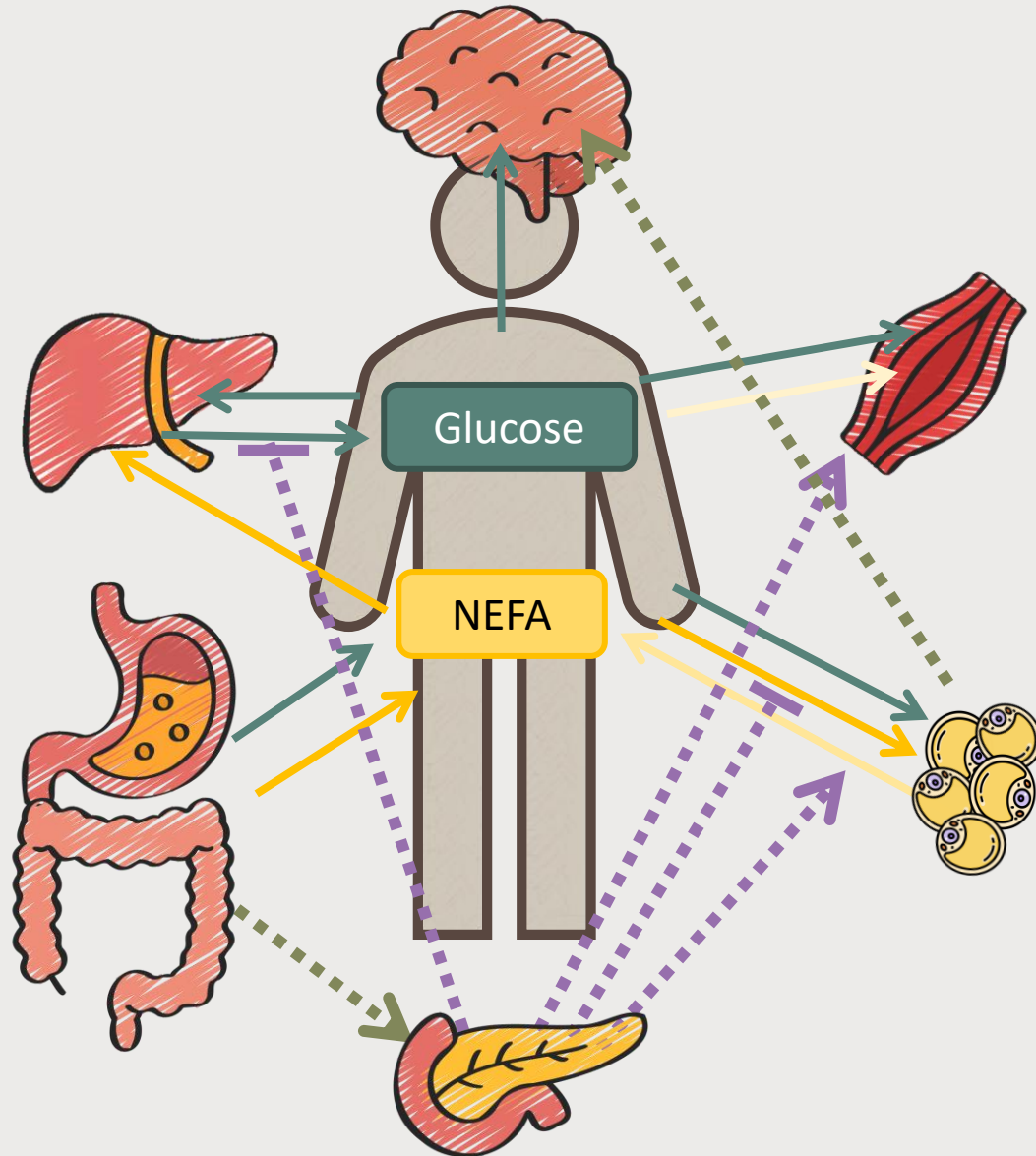
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METABOLIC RESILIENCE

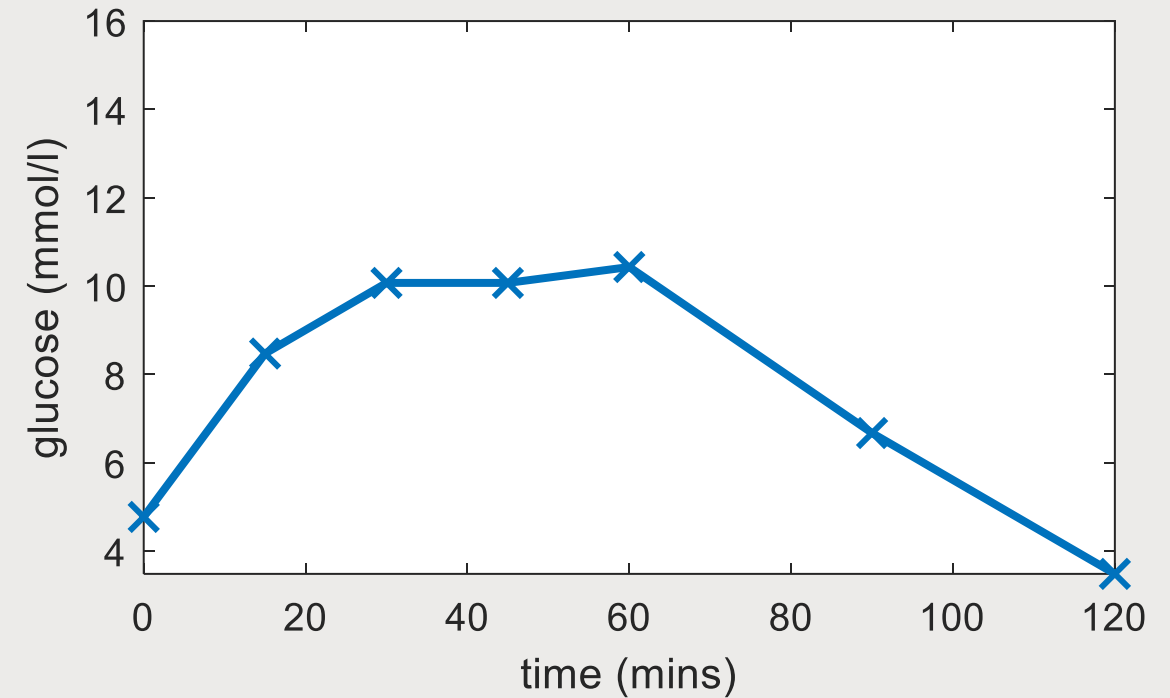


METABOLIC RESILIENCE

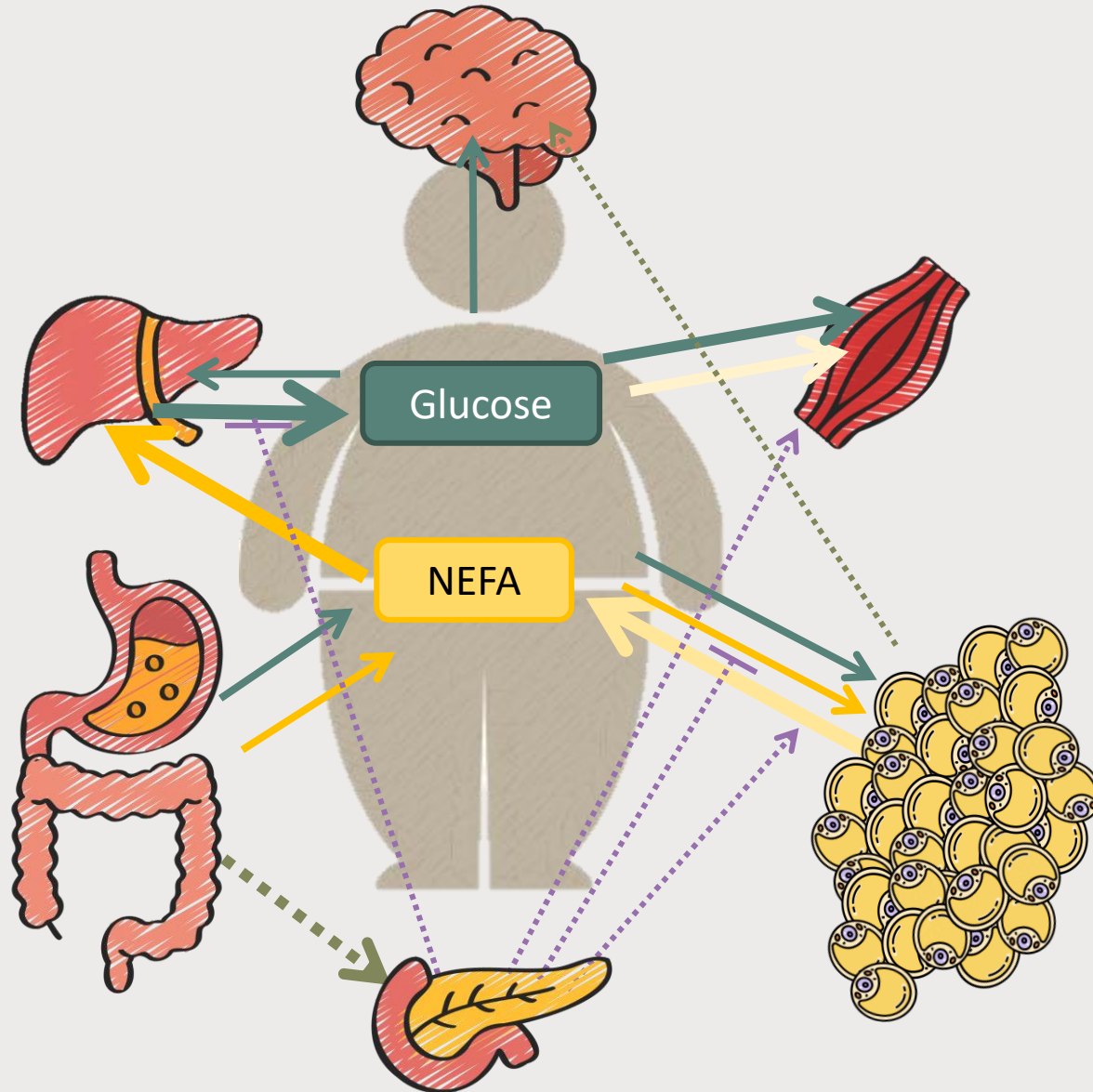


Healthy
individual

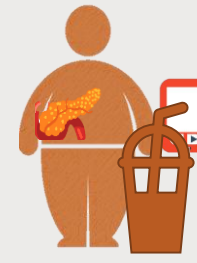
Meal Response



METABOLIC RESILIENCE

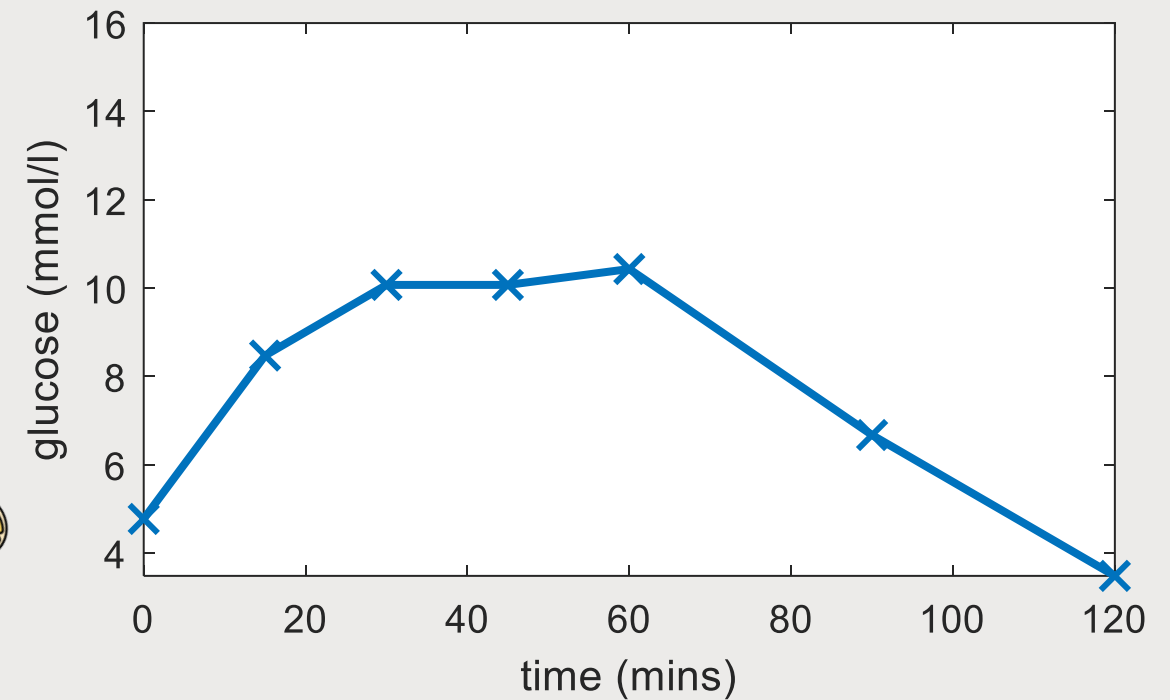


Healthy individual

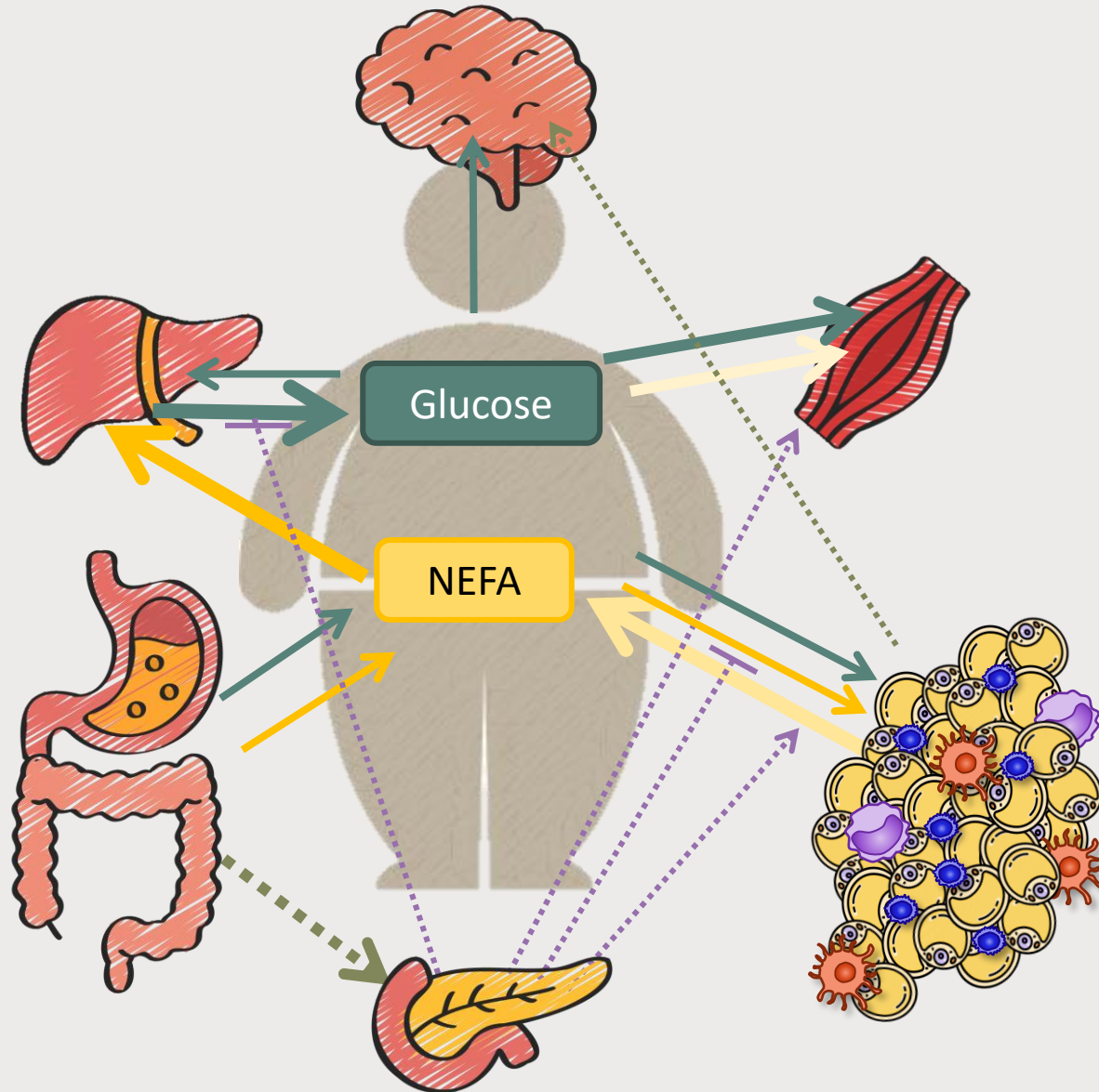


Individual with insulin resistance.

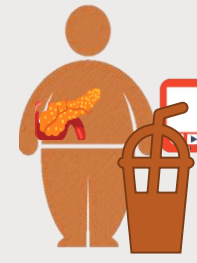
Meal Response



METABOLIC RESILIENCE

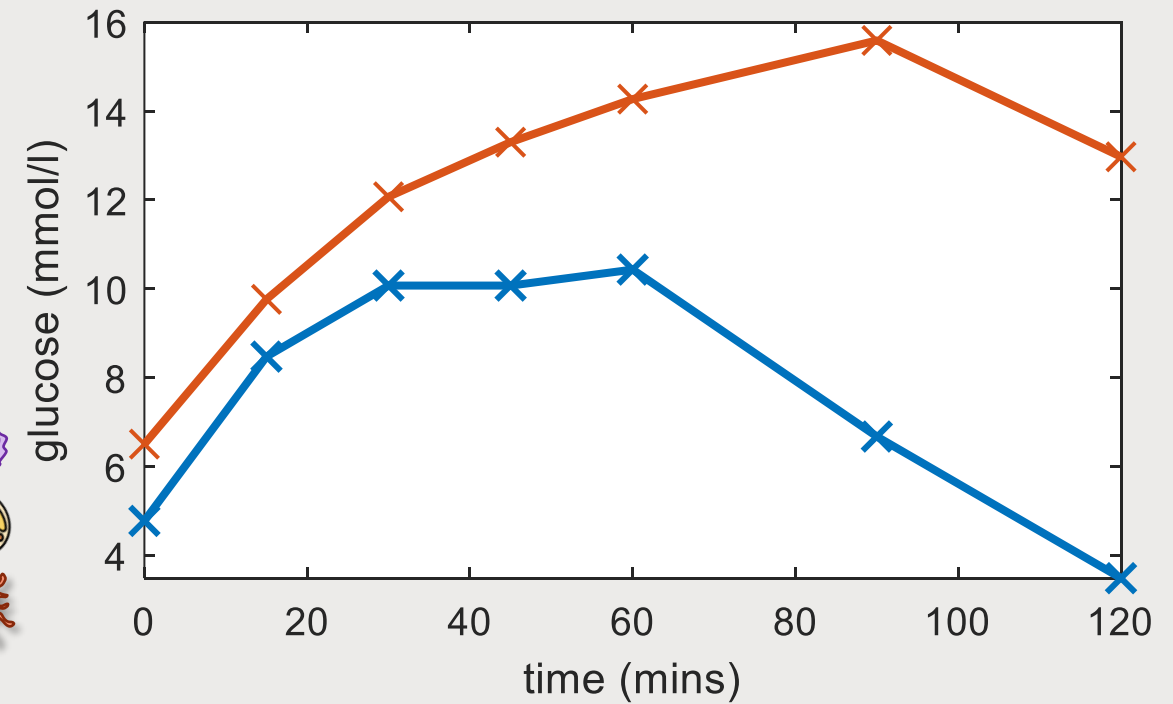


Healthy individual



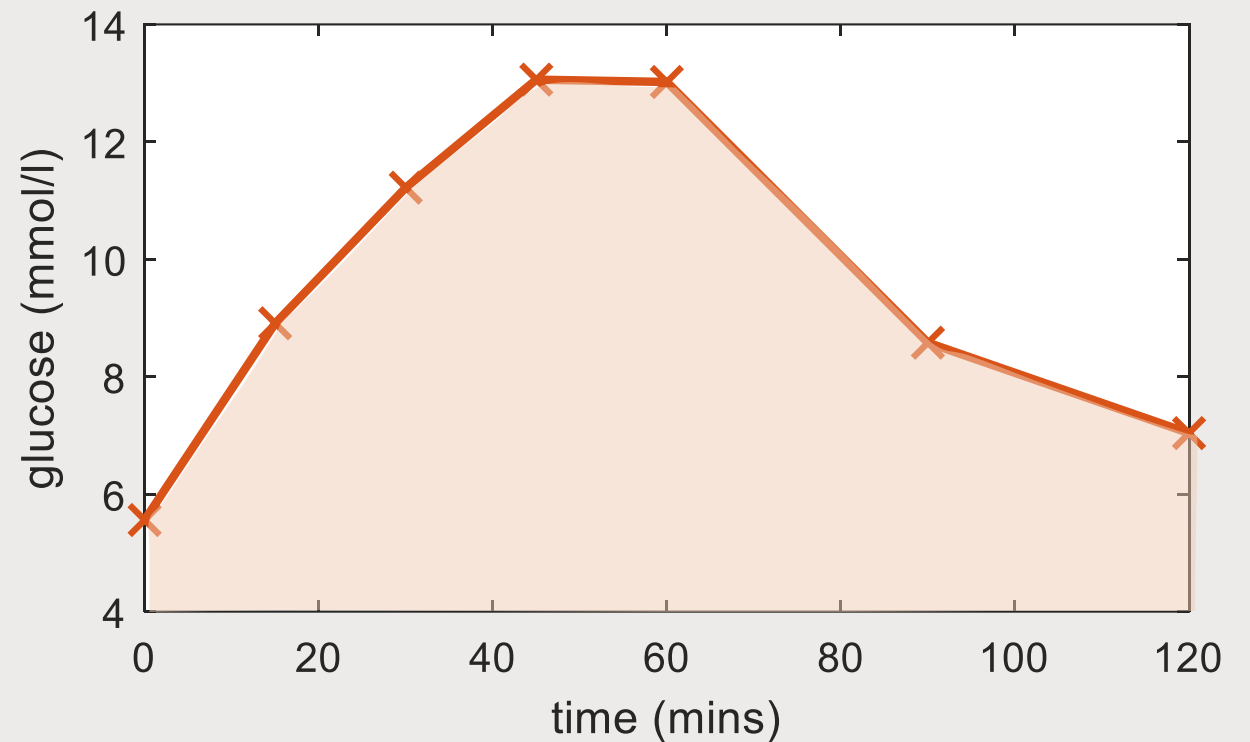
Individual with insulin resistance.

Meal Response



HOW TO QUANTIFY METABOLIC RESILIENCE

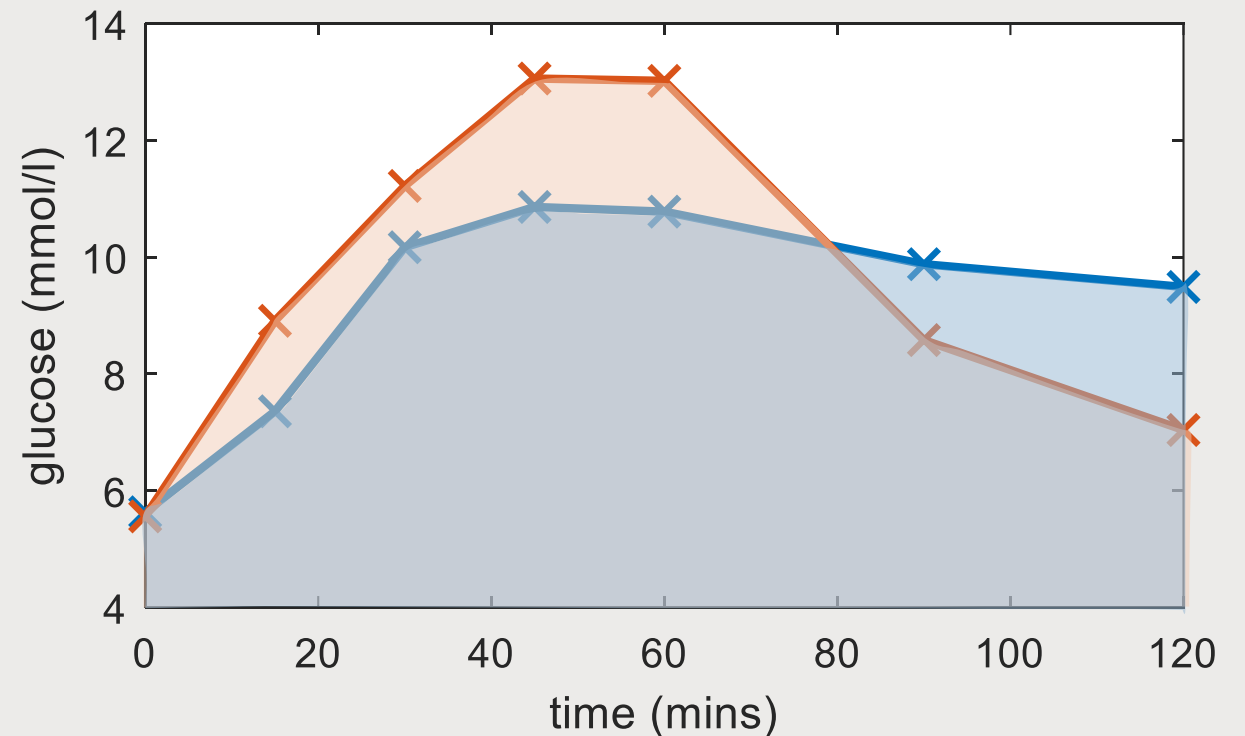
- Fasting values
- 2hour values
- Mean value
- Area under curve



HOW TO QUANTIFY METABOLIC RESILIENCE

- AUC reduces dynamic information to a single value.
- Two curves with very different dynamics may have very similar AUC values.

	Person 5	person16
AUC	1175	1169
Peak heigh	13.06	10.86
Glucose 120	7.04	9.49
Slope decline	-0.09	-0.02

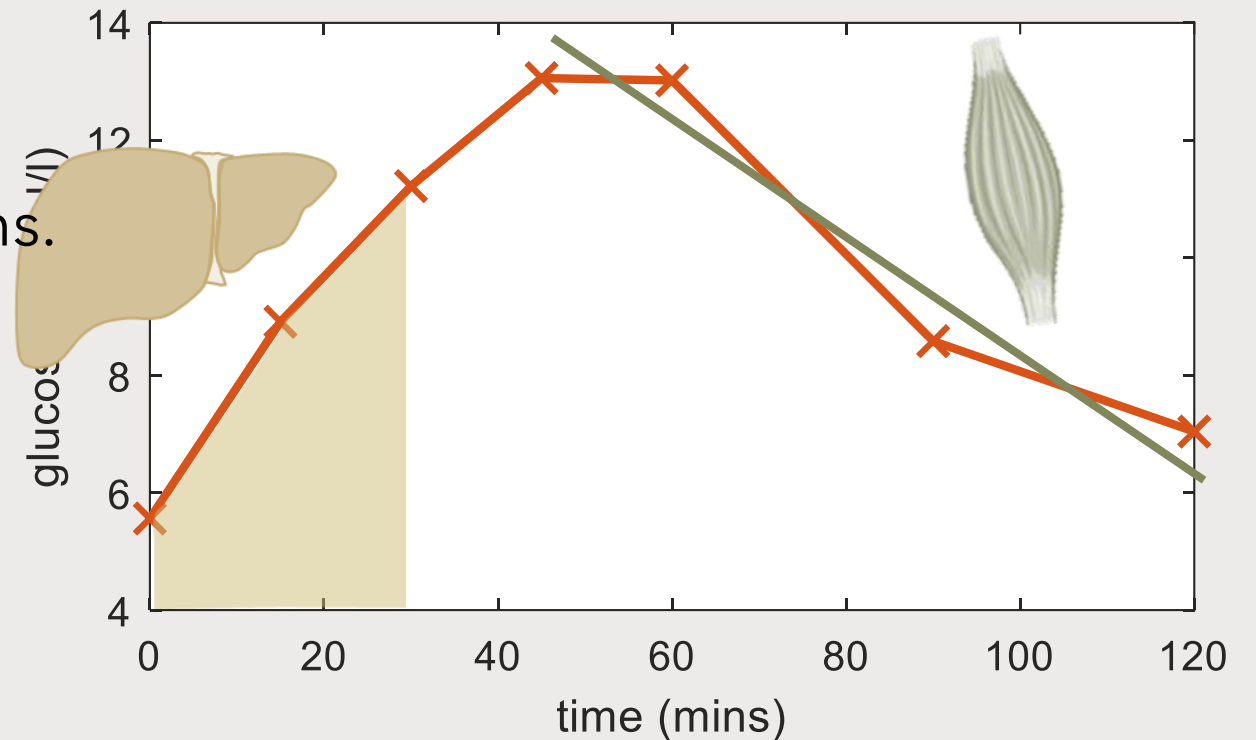


HOW TO QUANTIFY METABOLIC RESILIENCE

Different parts of the meal response curve tell us something about the health of individual tissues and organs.

Early phase of glucose response is governed by liver insulin sensitivity.

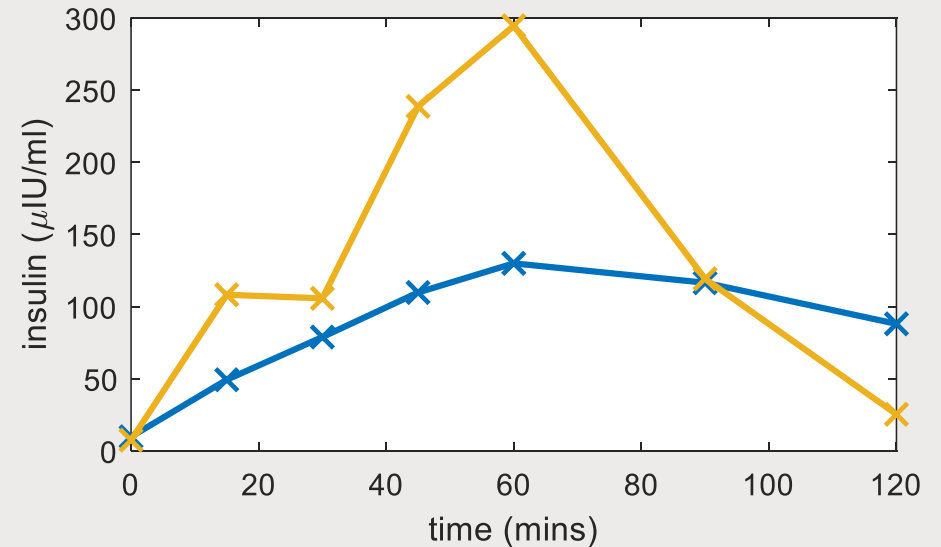
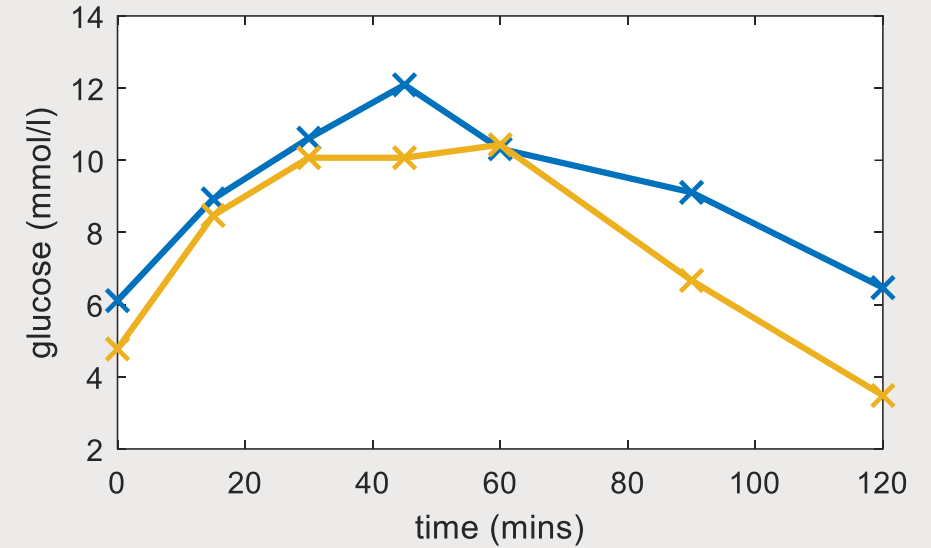
Later phase of glucose response is indicative of skeletal muscle health.



HOW TO QUANTIFY METABOLIC RESILIENCE

The relative plasma levels of **multiple metabolites and hormones** may be needed to provide a complete picture of metabolic resilience.

The yellow individual requires much higher insulin levels to maintain glycemic control, suggesting they have insulin resistance.



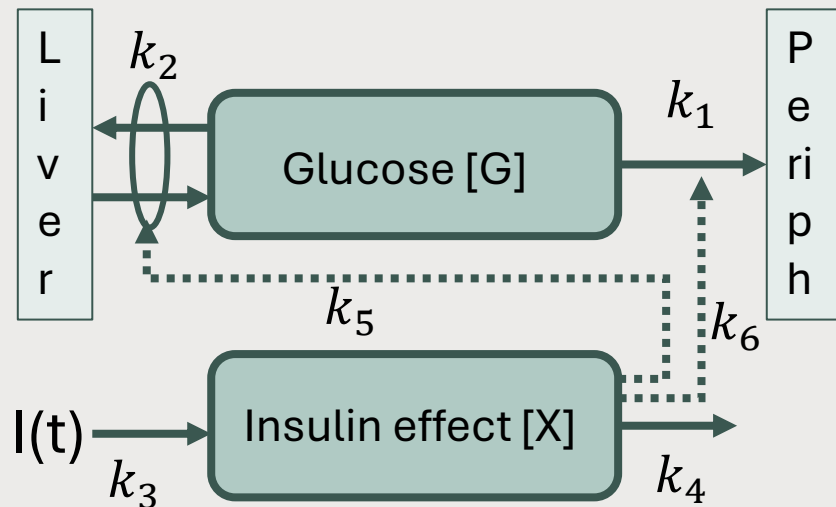
MATHEMATICAL MODELS OF GLUCOSE-INSULIN SYSTEM

IDENTIFICATION OF A MINIMAL MODEL OF GLUCOSE DISAPPEARANCE FOR ESTIMATING INSULIN SENSITIVITY

R. N. Bergman*, G. Bortolan**, C. Cobelli** and G. Toffolo**

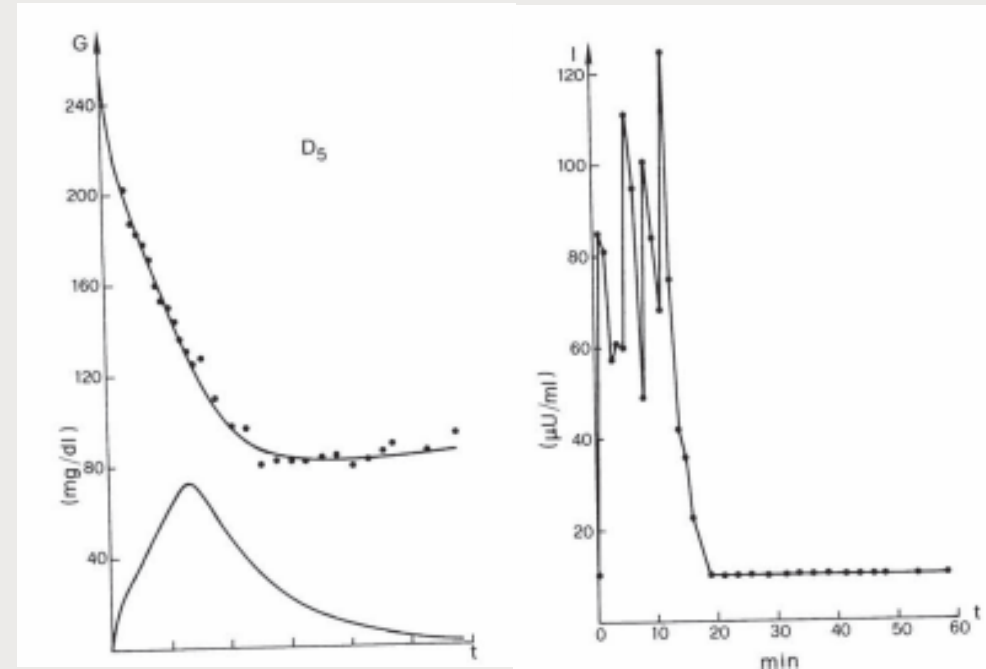
*Biomedical Engineering Center, The Technological Institute, Northwestern University, Evanston, Illinois 60201, U.S.A.

**Laboratorio per Ricerche di Dinamica dei Sistemi e di Bioingegneria, Consiglio Nazionale delle Ricerche, Padova, Italy and Istituto di Elettrotecnica e di Elettronica, Università di Padova, Padova, Italy

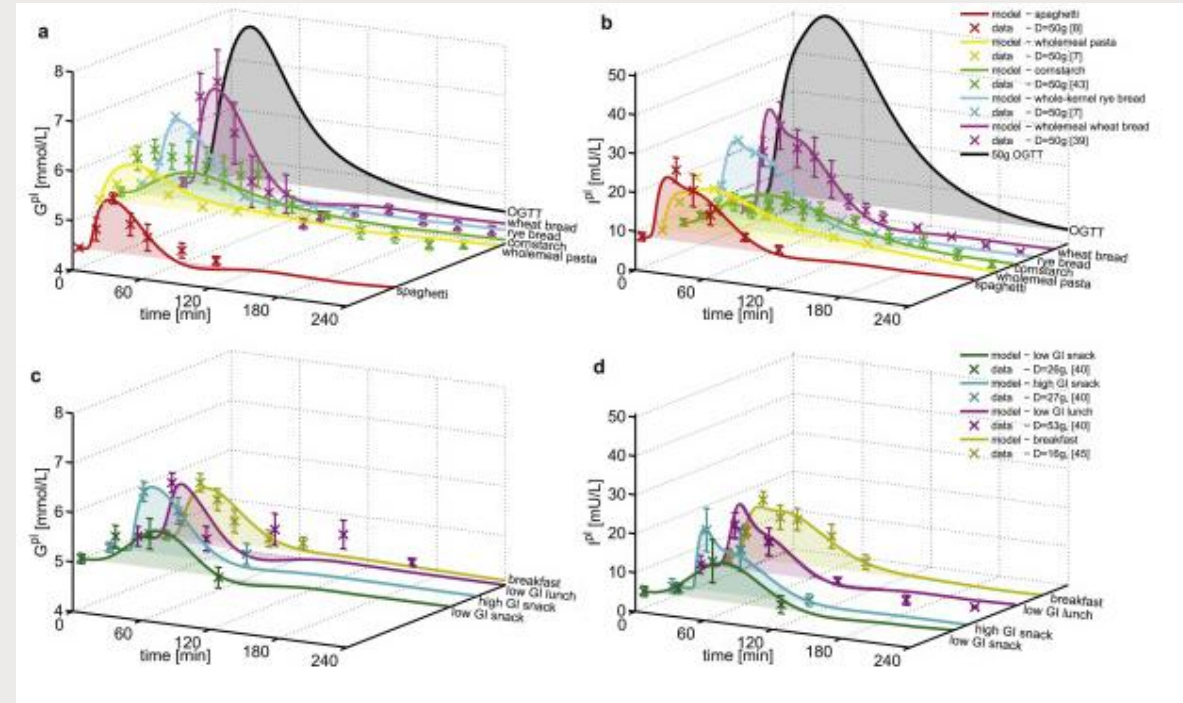
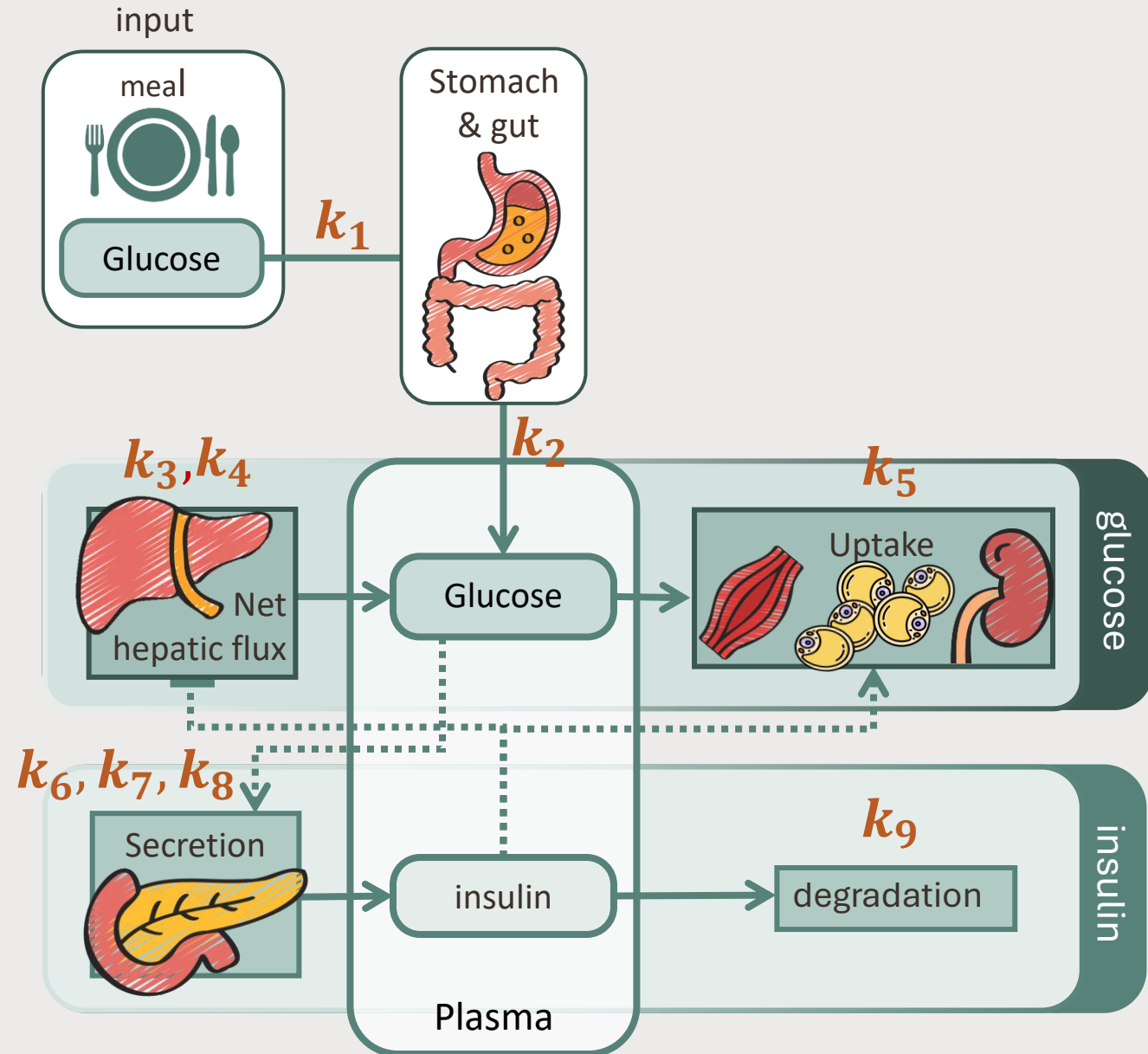


$$\frac{d[G]}{dt} = k_2 \cdot k_5[X] - (k_1 \cdot k_6[X] + k_2 \cdot k_5[X])[G]$$

$$\frac{d[X]}{dt} = k_3[I] - k_4[X]$$



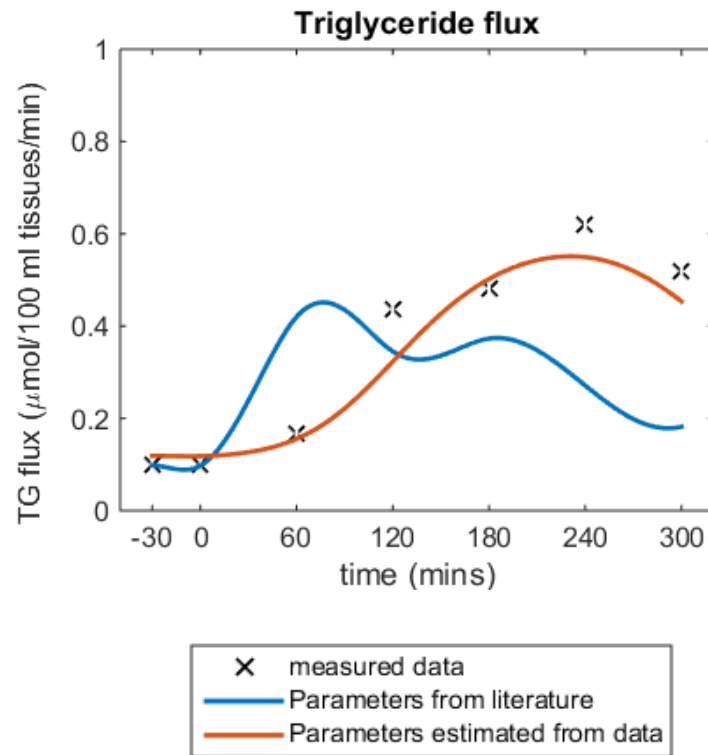
EINDHOVEN DIABETES EDUCATION SIMULATOR



Maas AH, Rozendaal YJ et al. A physiology-based model describing heterogeneity in glucose metabolism: the core of the Eindhoven Diabetes Education Simulator (E-DES). *J Diabetes Sci Technol*. 2015 Mar;9(2):282-92. doi: 10.1177/1932296814562607.

MAXIMUM LIKELIHOOD ESTIMATION

$$\frac{d[TG_{PL}]}{dt} = 0.0096 [I_{LPL}] [TG_{PL}]$$



Model can be described as

$$y(t) = f(u(t), \theta) + \xi(t)$$

residuals

$$\Rightarrow \xi(t) = y(t) - f(u(t), \theta)$$

Probability density function

$$p(\xi(t)) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\xi(t)^2 / (2\sigma^2)}$$

Maximise the likelihood

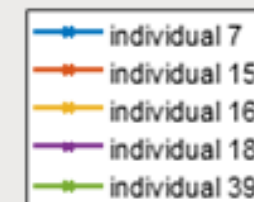
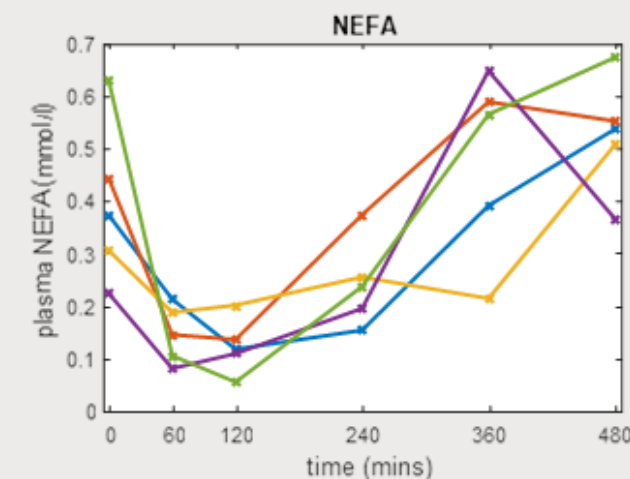
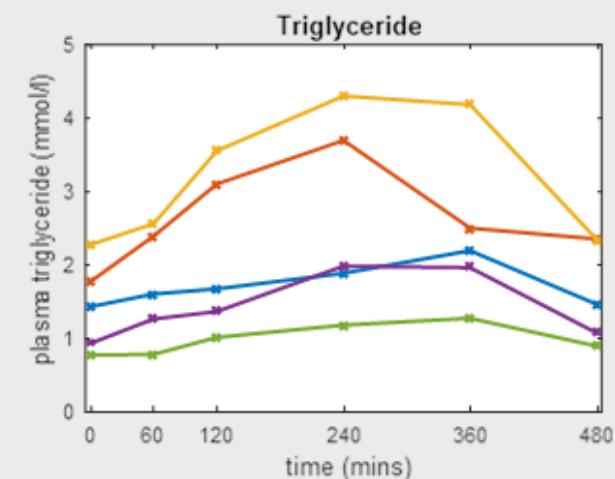
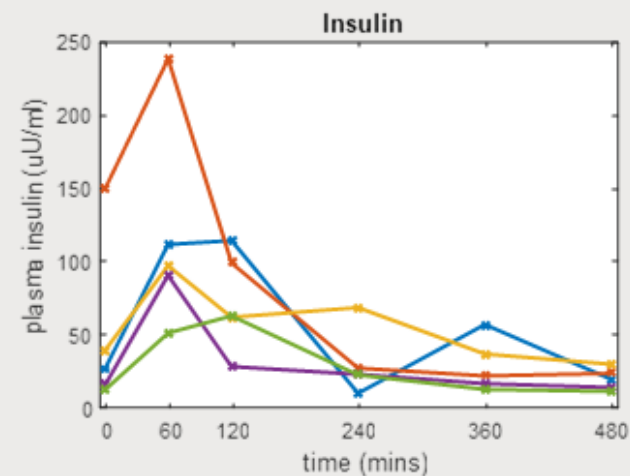
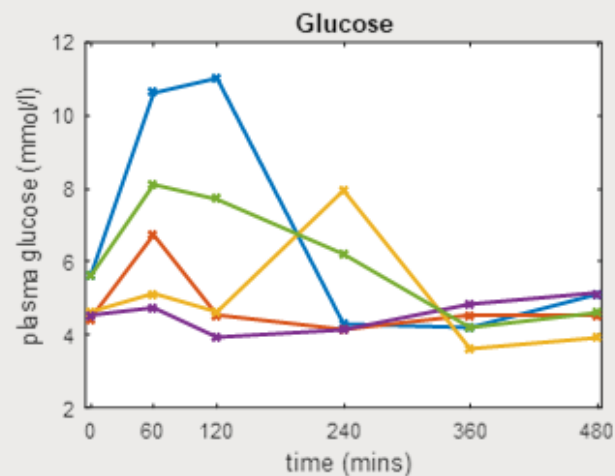
$$\hat{\theta} = \max_{\theta} L(y(t)|\theta)$$

Capturing metabolic resilience

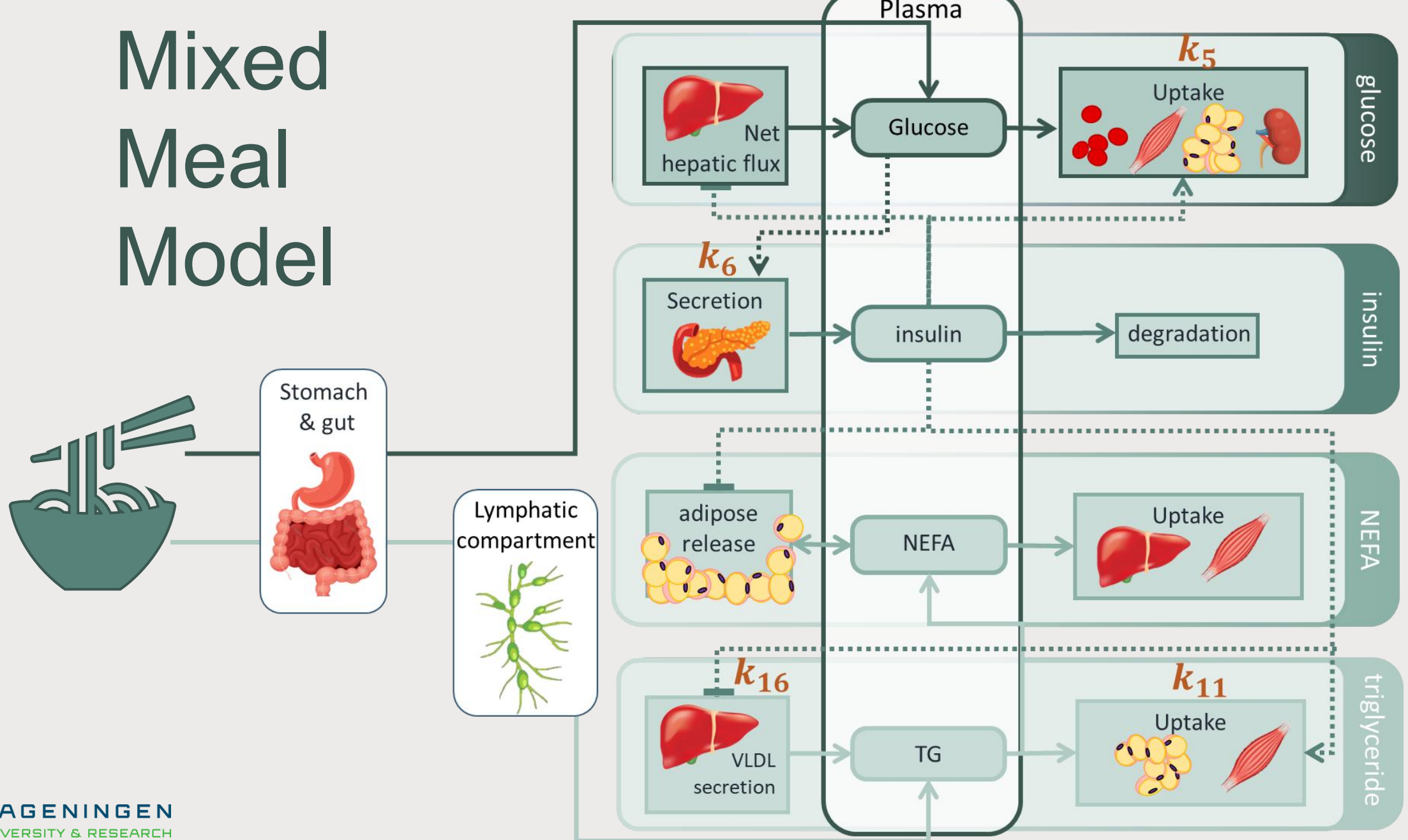


Meal challenge test

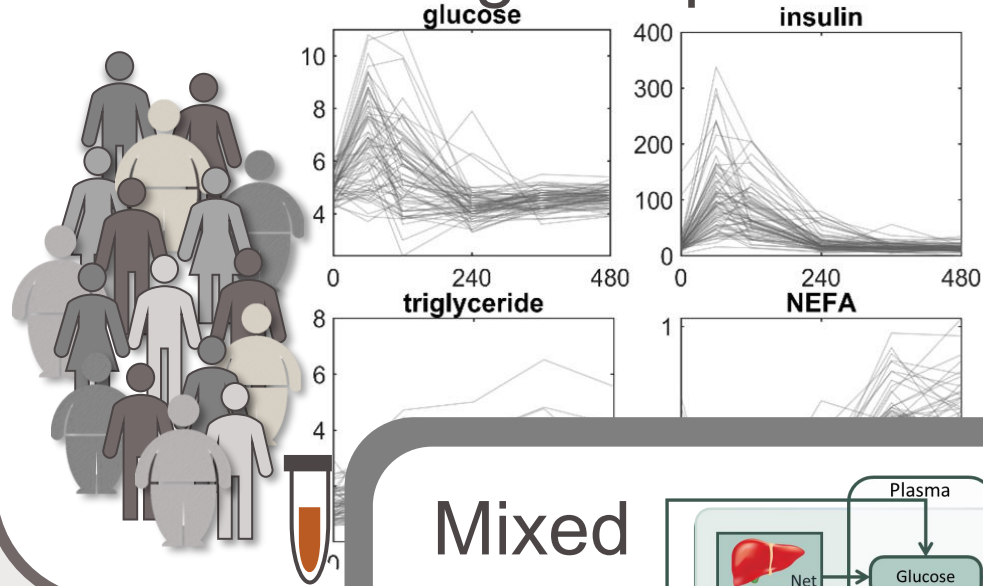
- 75g glucose
- 60g lipid
- 20g protein



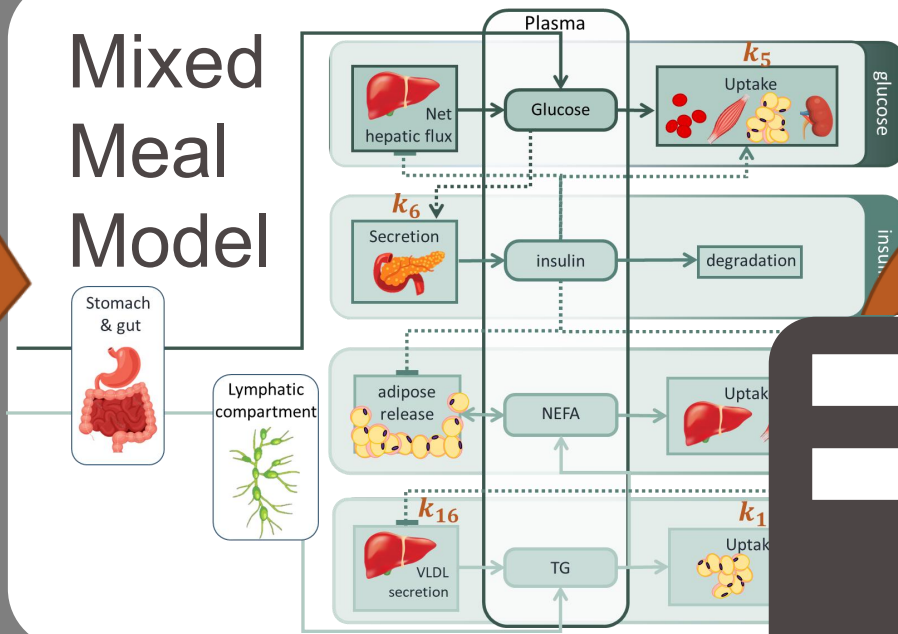
Mixed Meal Model



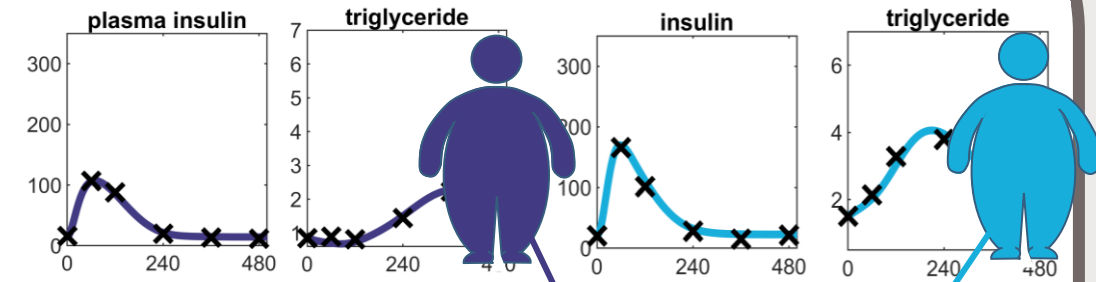
Meal challenge responses



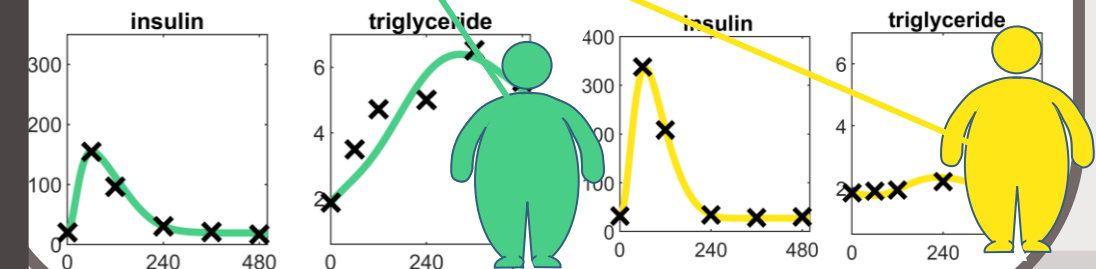
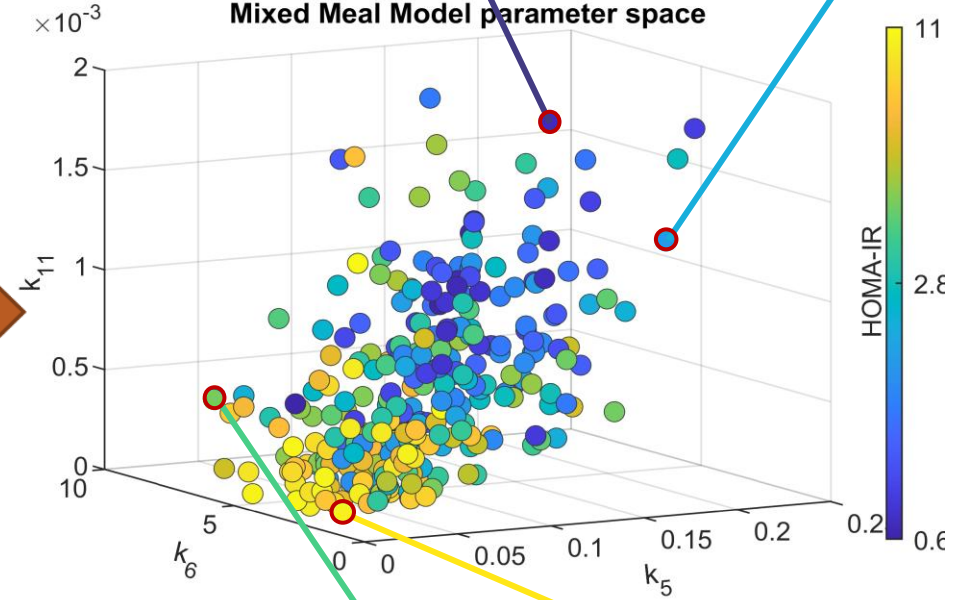
Mixed Meal Model



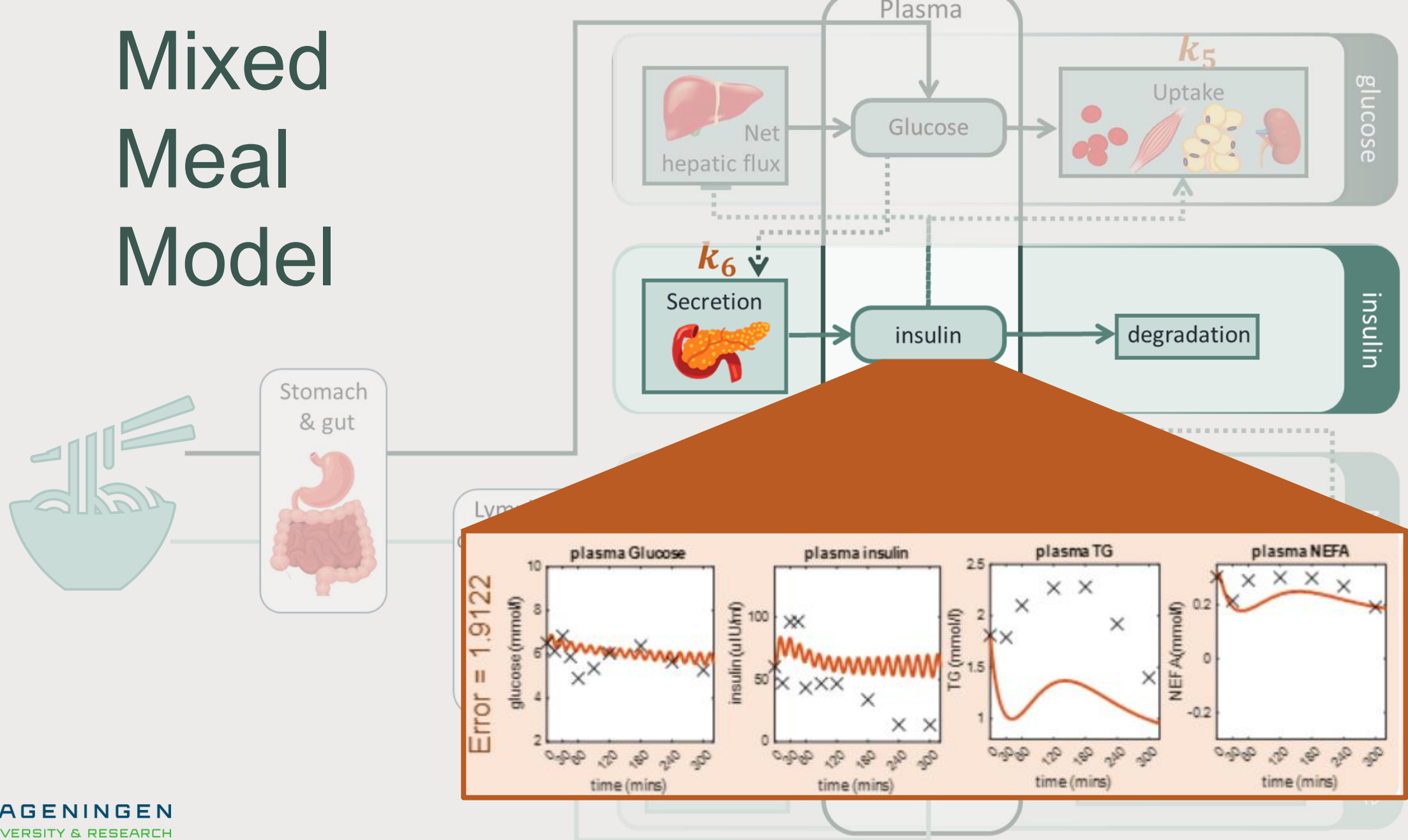
Metabolic resilience phenotype



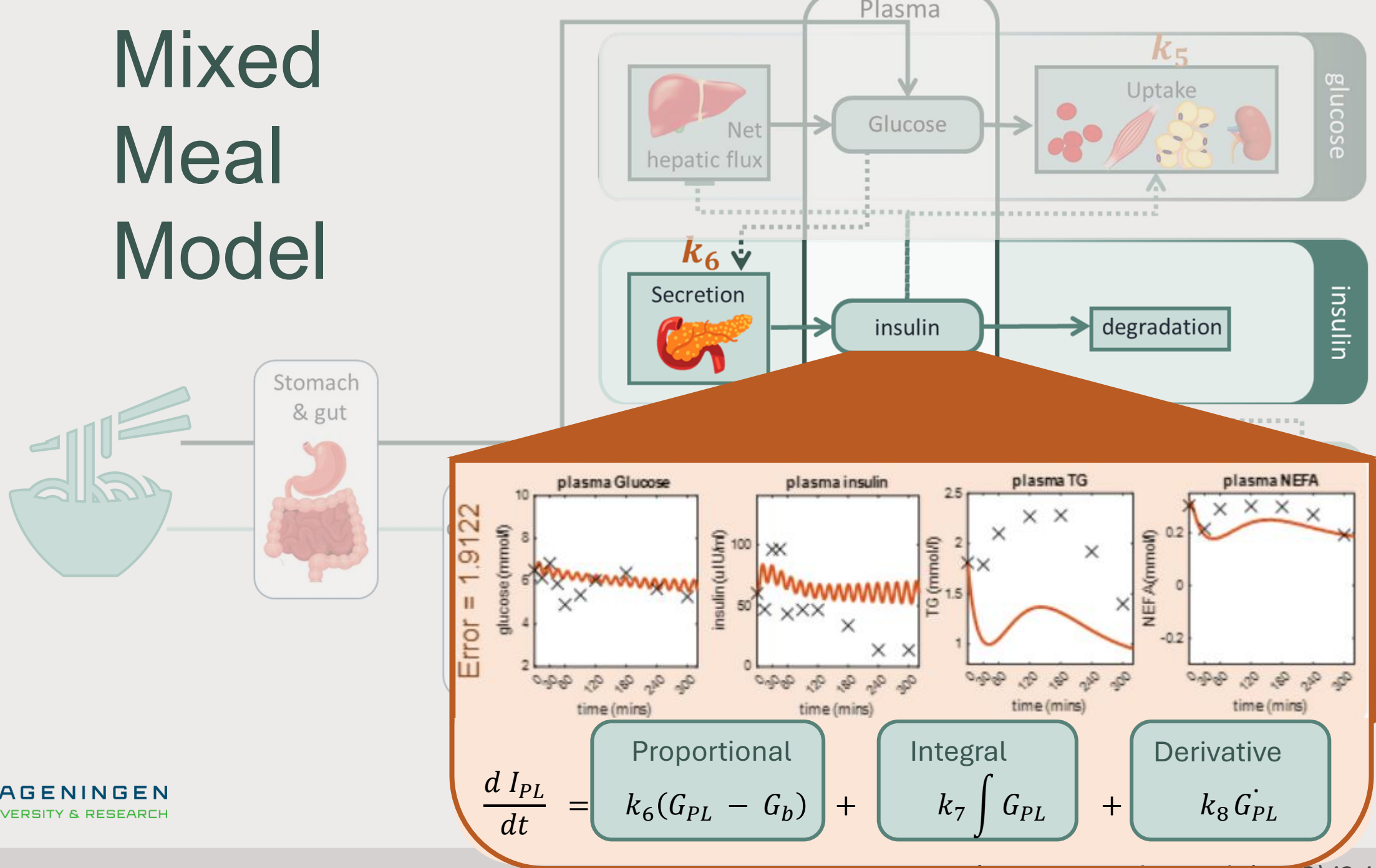
Mixed Meal Model parameter space



Mixed Meal Model



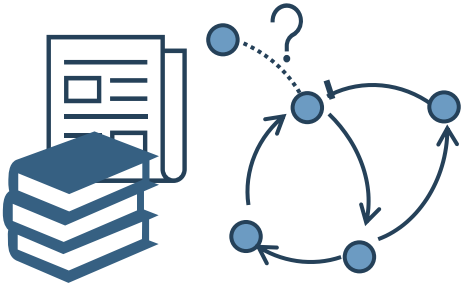
Mixed Meal Model



Systems Approach

Bottom-up

Knowledge



Biological interactions are identified from literature

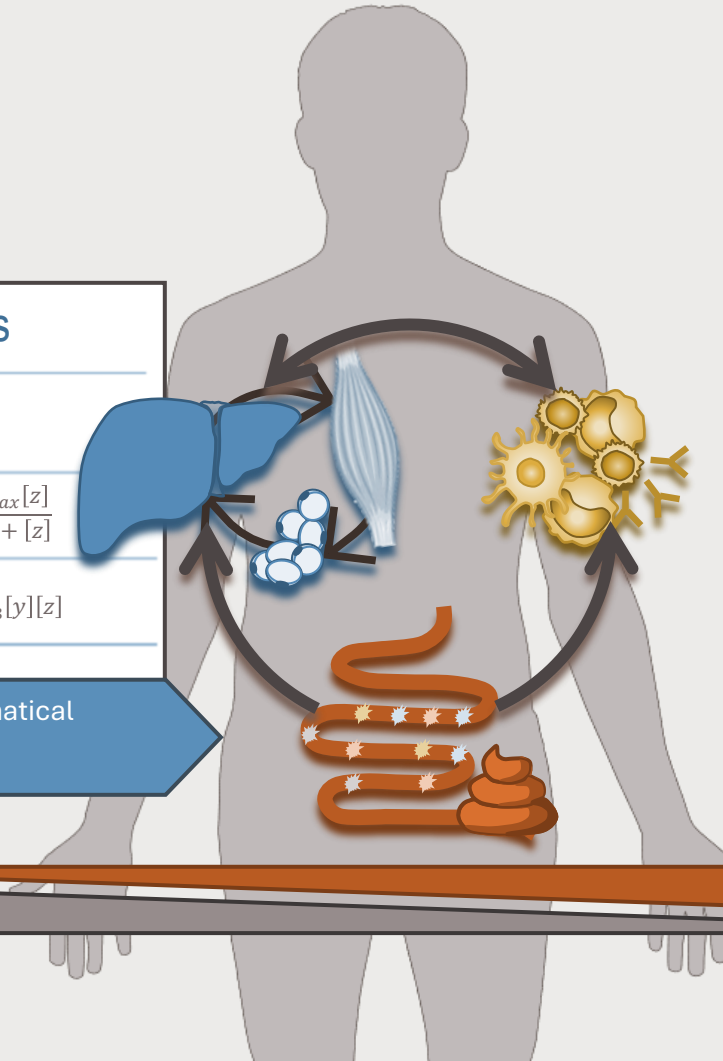
Model equations

$$\frac{d[x]}{dt} = k_1[y] - \frac{V_{max}[x]}{k_2 + [x]}$$

$$\frac{d[z]}{dt} = k_3[y][z] + k_4 - \frac{V_{max}[z]}{k_2 + [z]}$$

$$\frac{d[y]}{dt} = \frac{V_{max}[z]}{k_2 + [z]} - k_1[y] - k_3[y][z]$$

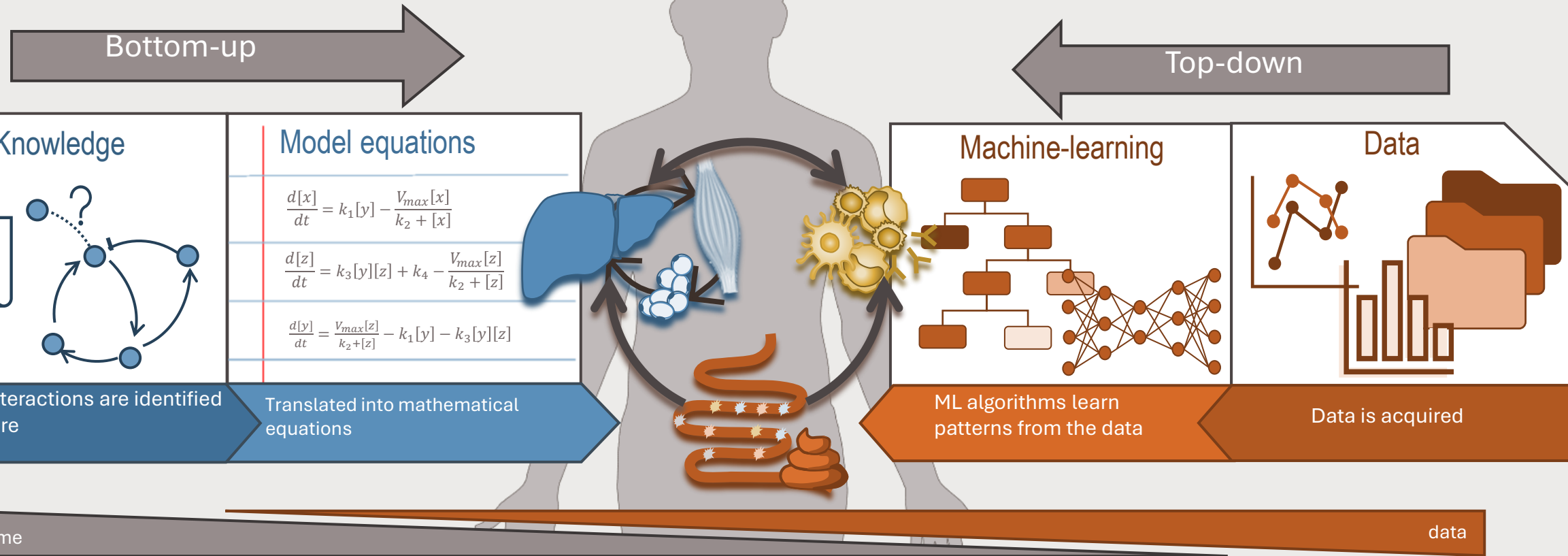
Translated into mathematical equations



time

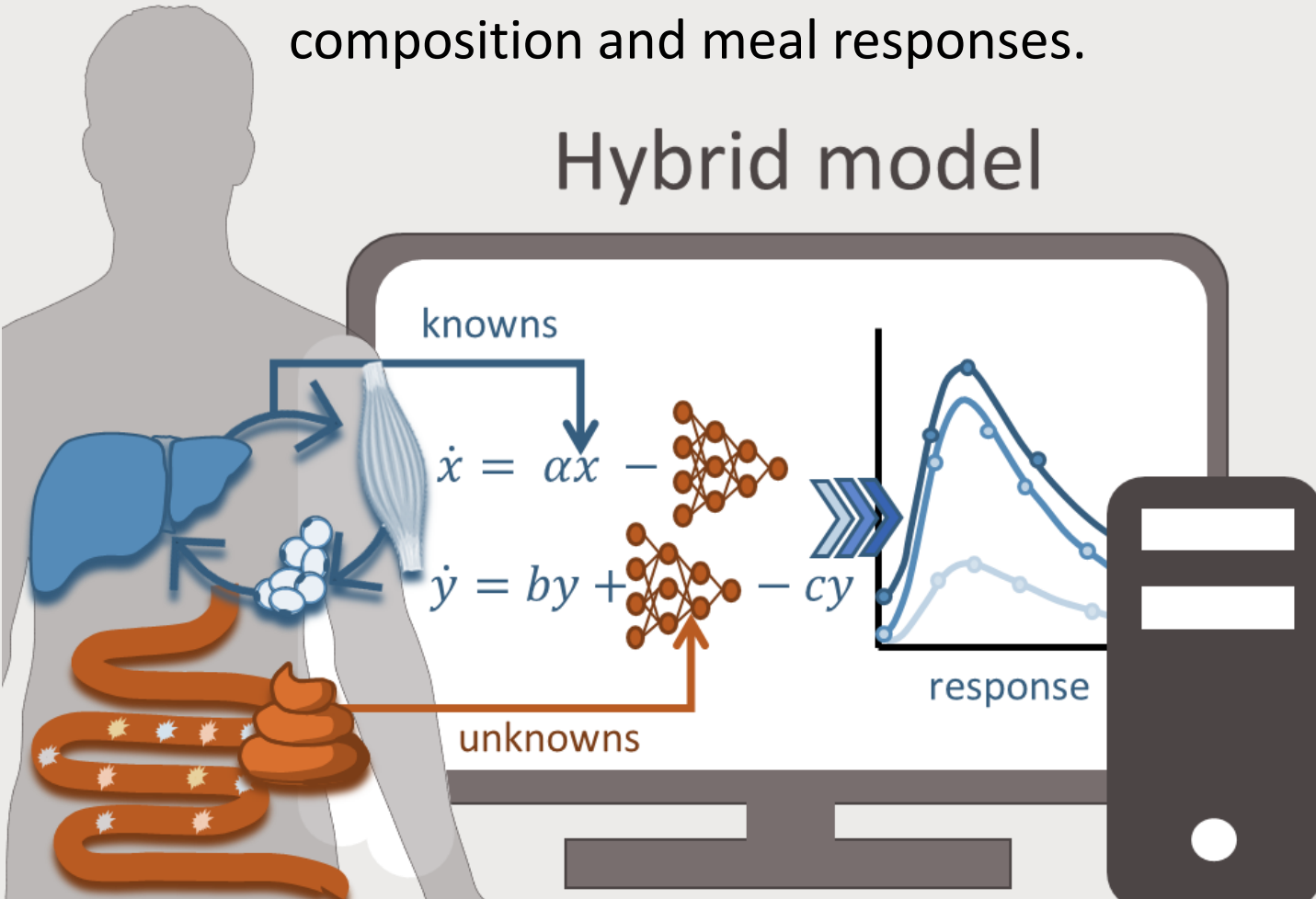
data

Systems Approach



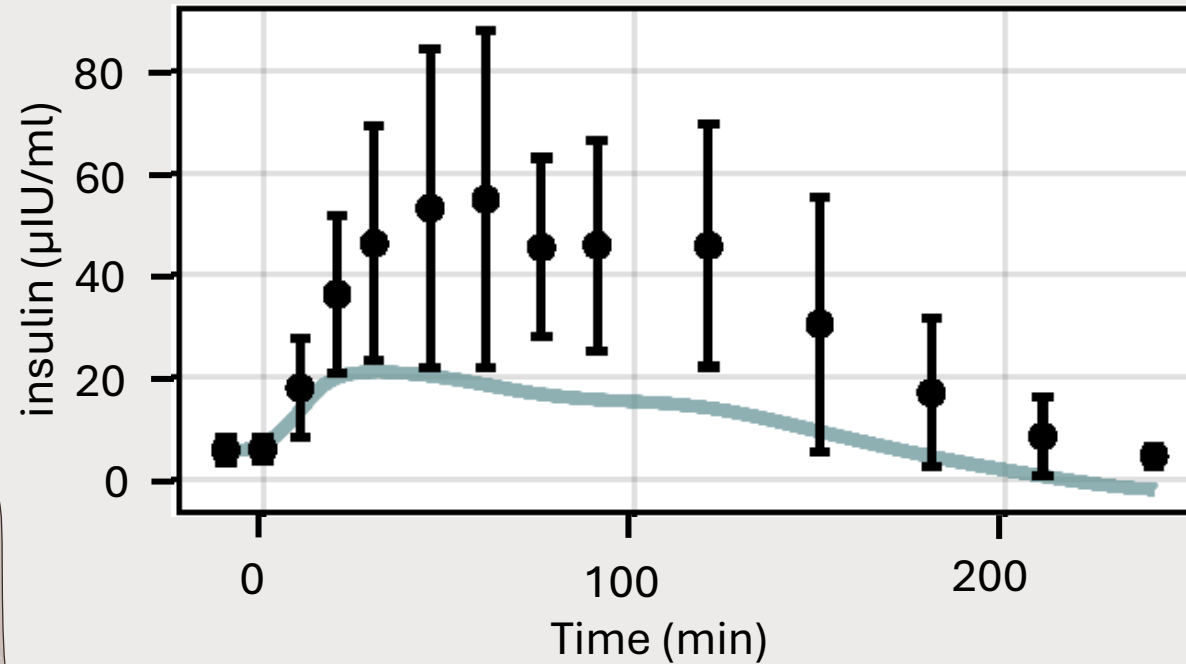
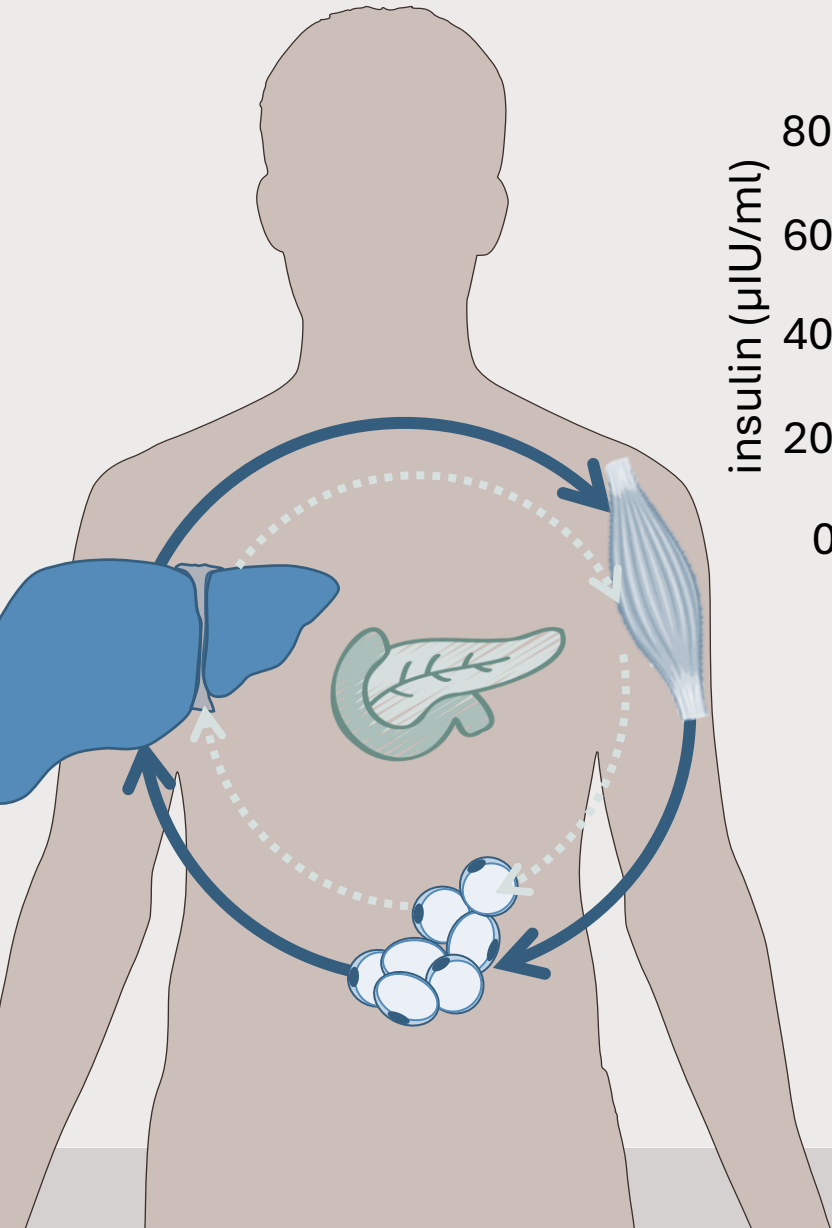
Hybrid-models

Combining **existing knowledge** with **artificial intelligence** to learn previously **unknown** relationships between high-dimension features such as microbiome composition and meal responses.



- Model can be trained with **fewer learning examples**.
- Facilitate multi-modal **data integration**.
- Facilitate **rapid model extension**.

Universal Differential Equations in Biology



— EDeS

$$\frac{d I_{PL}}{dt} =$$

Proportional

$$k_6 (G_{PL} - G_b)$$

+

Integral

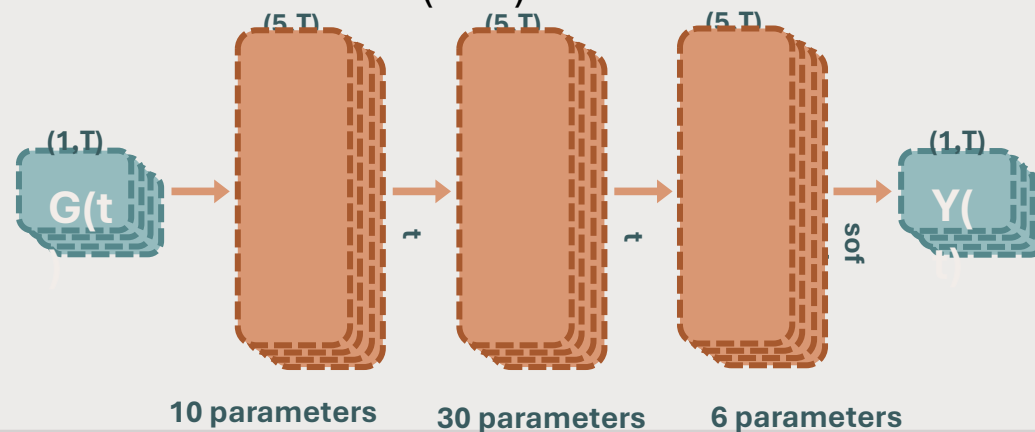
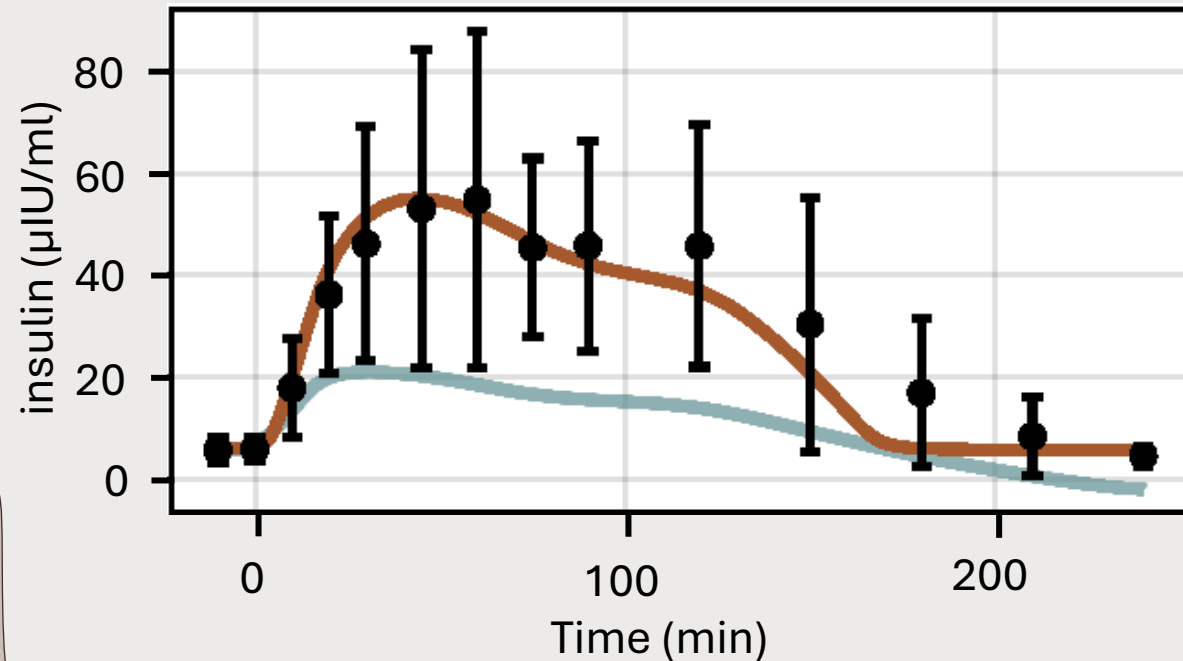
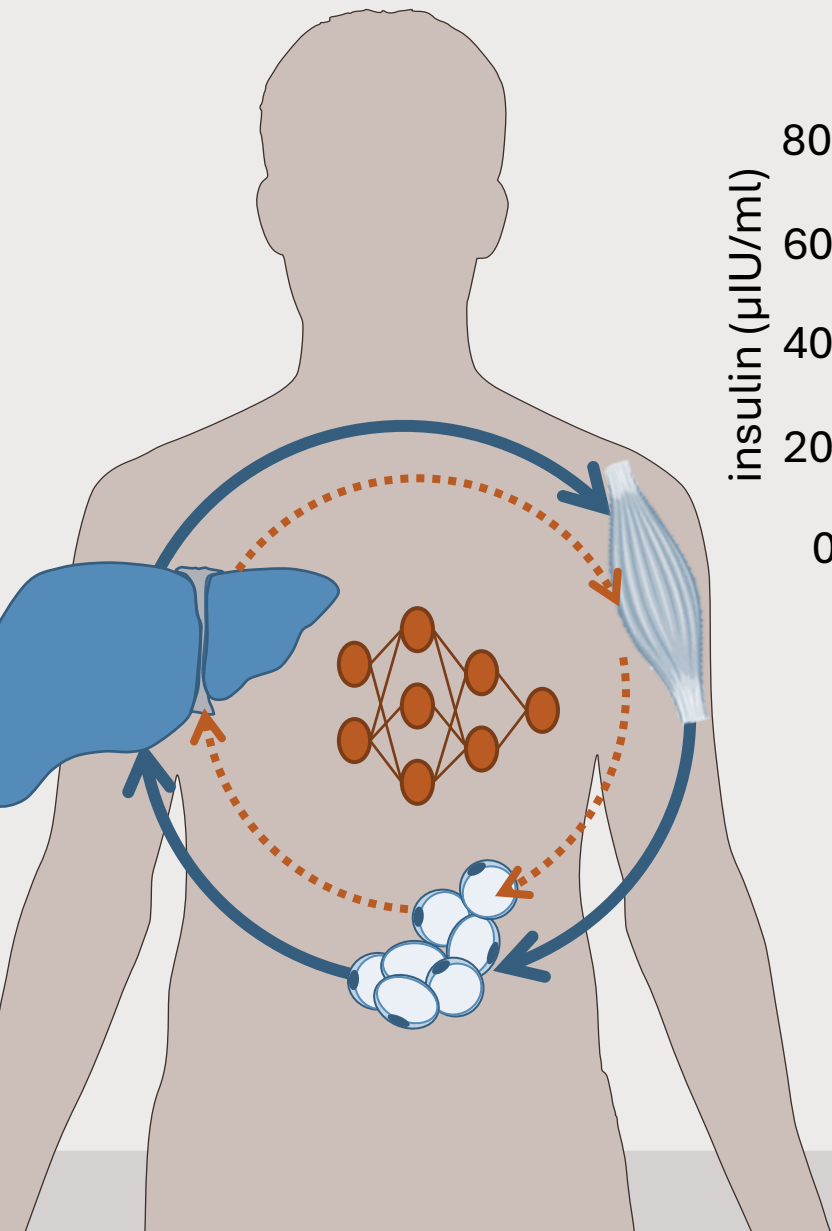
$$k_7 \int G_{PL}$$

+

Derivative

$$k_8 \dot{G}_{PL}$$

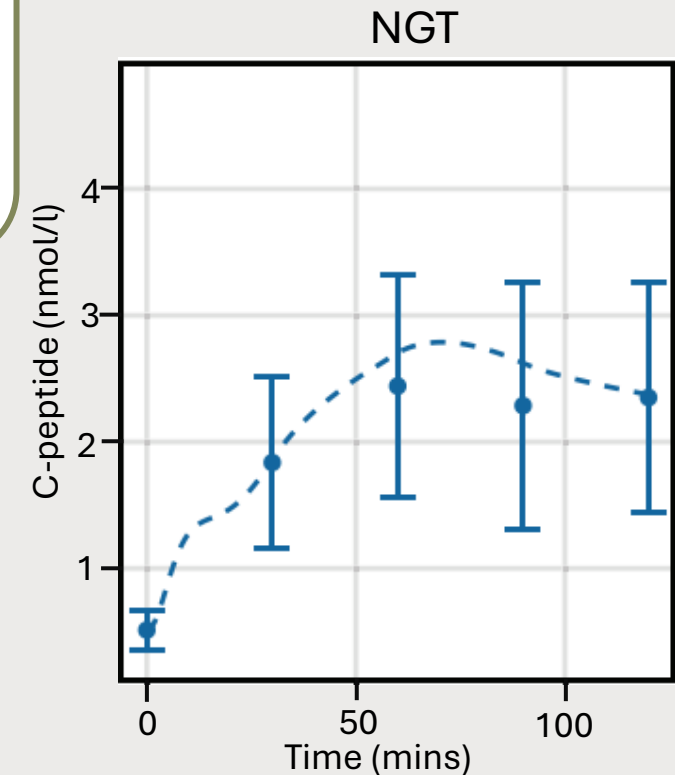
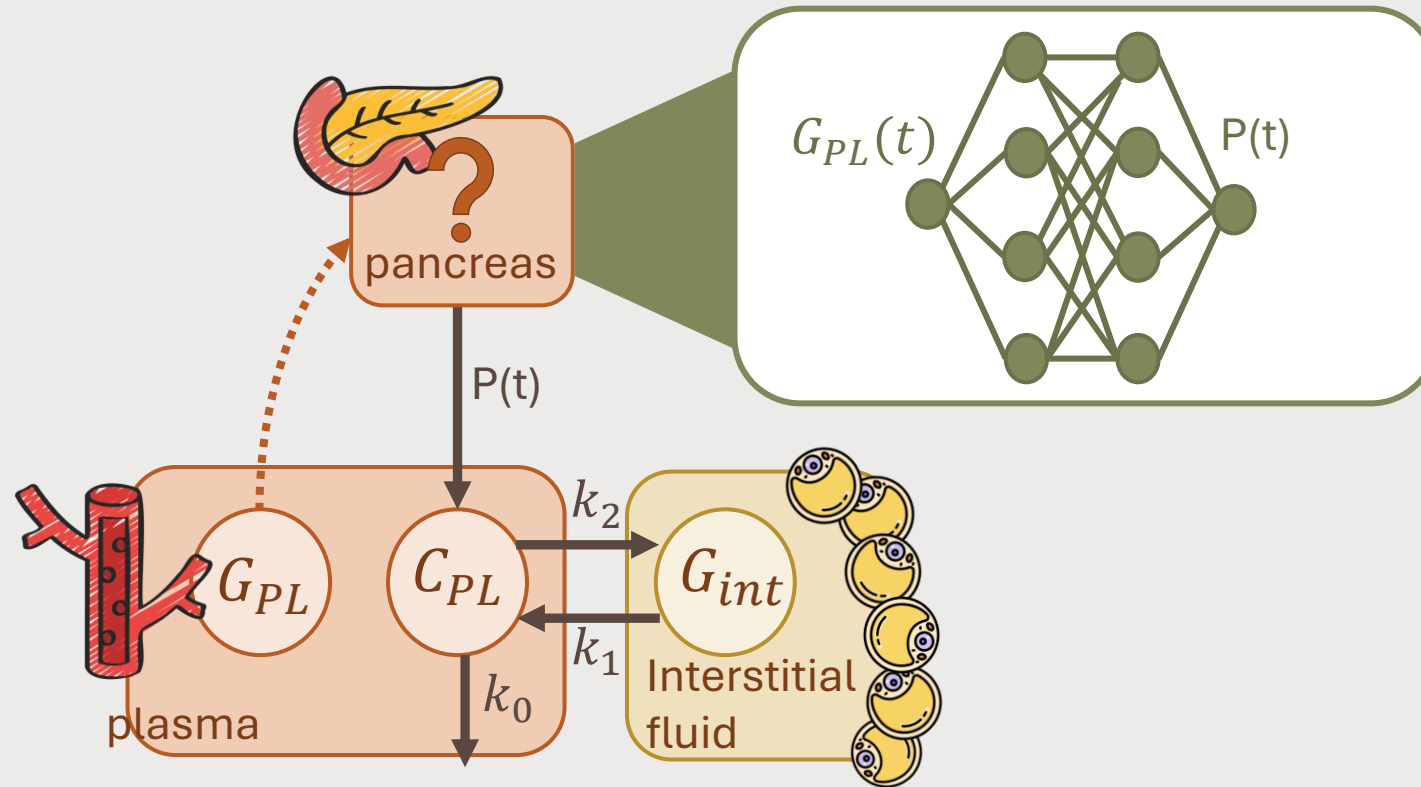
Universal Differential Equations in Biology



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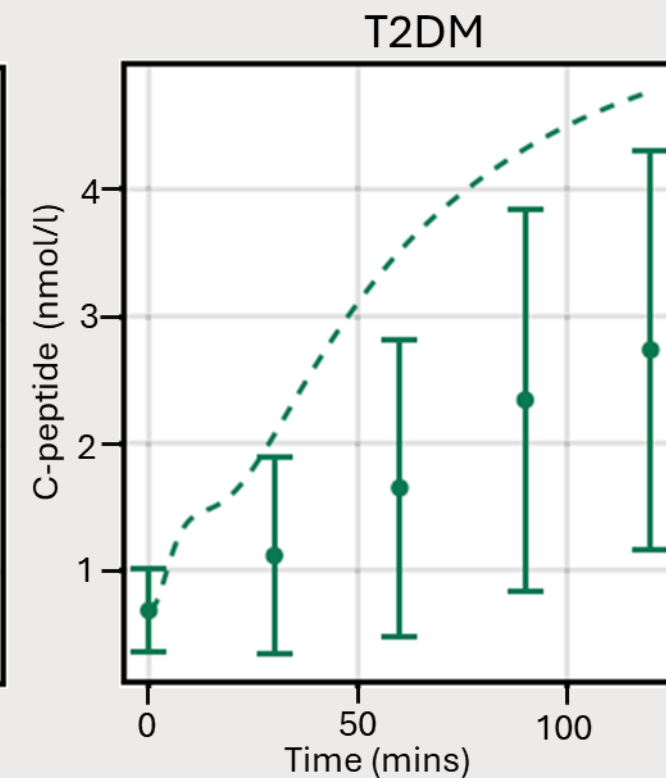
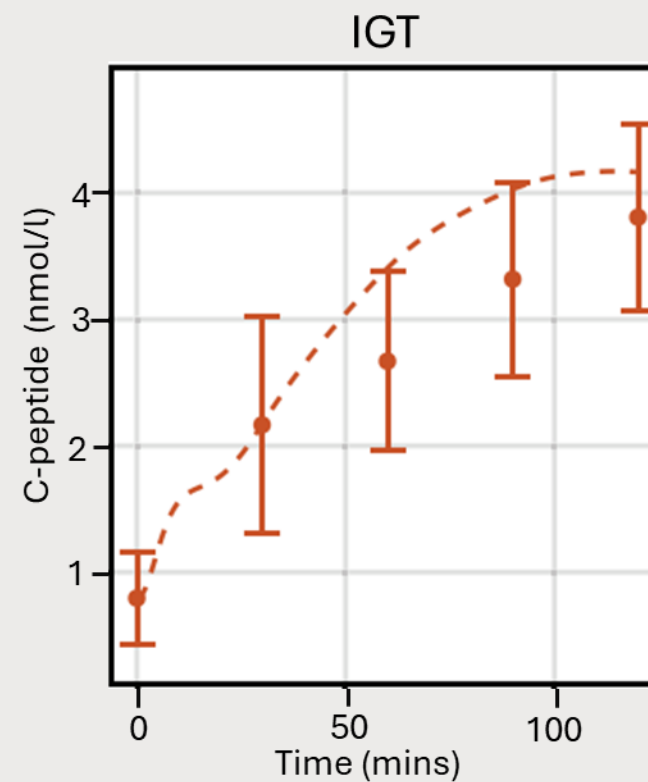
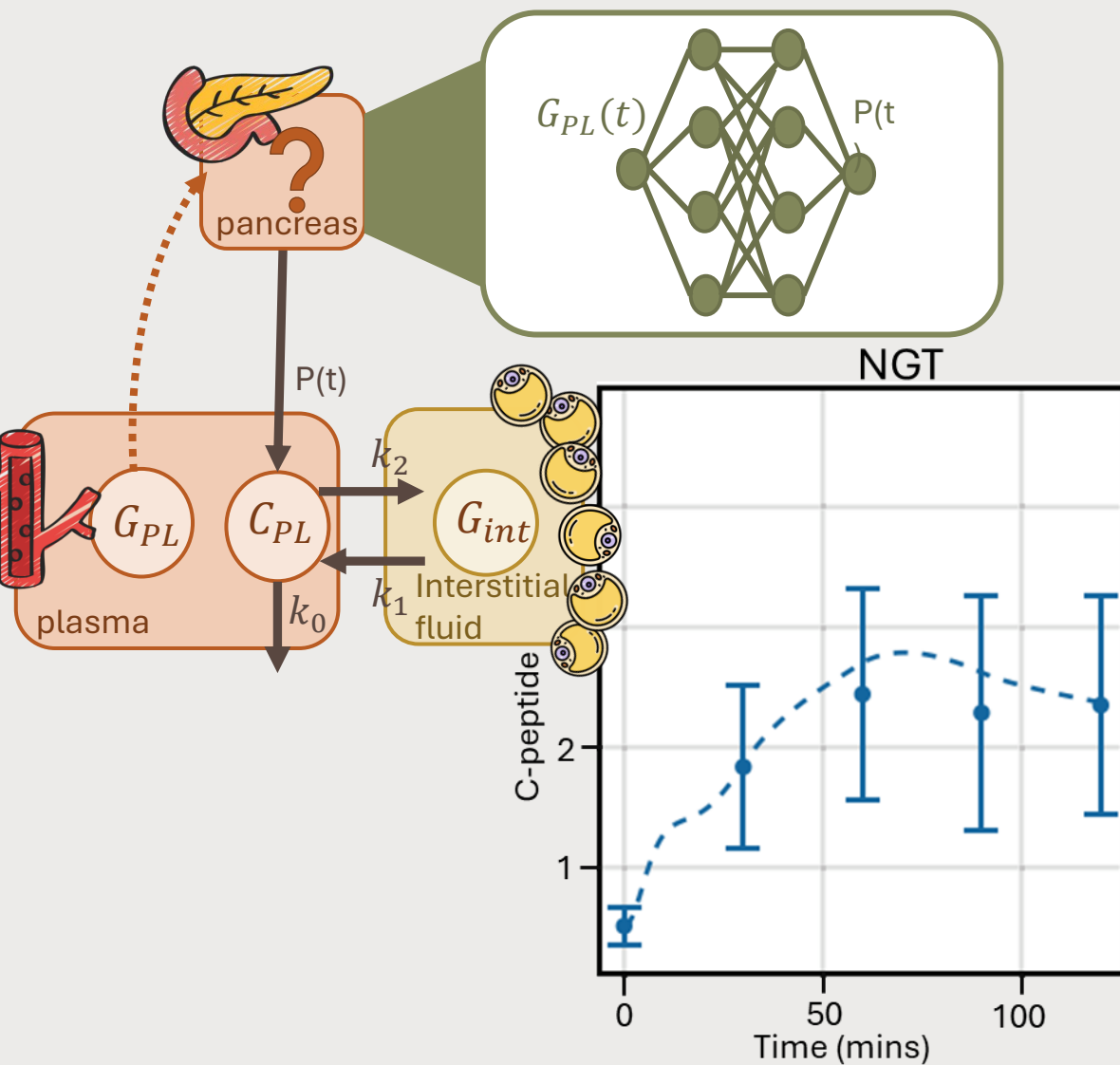
Max de Rooij



Universal Differential Equations in Biology



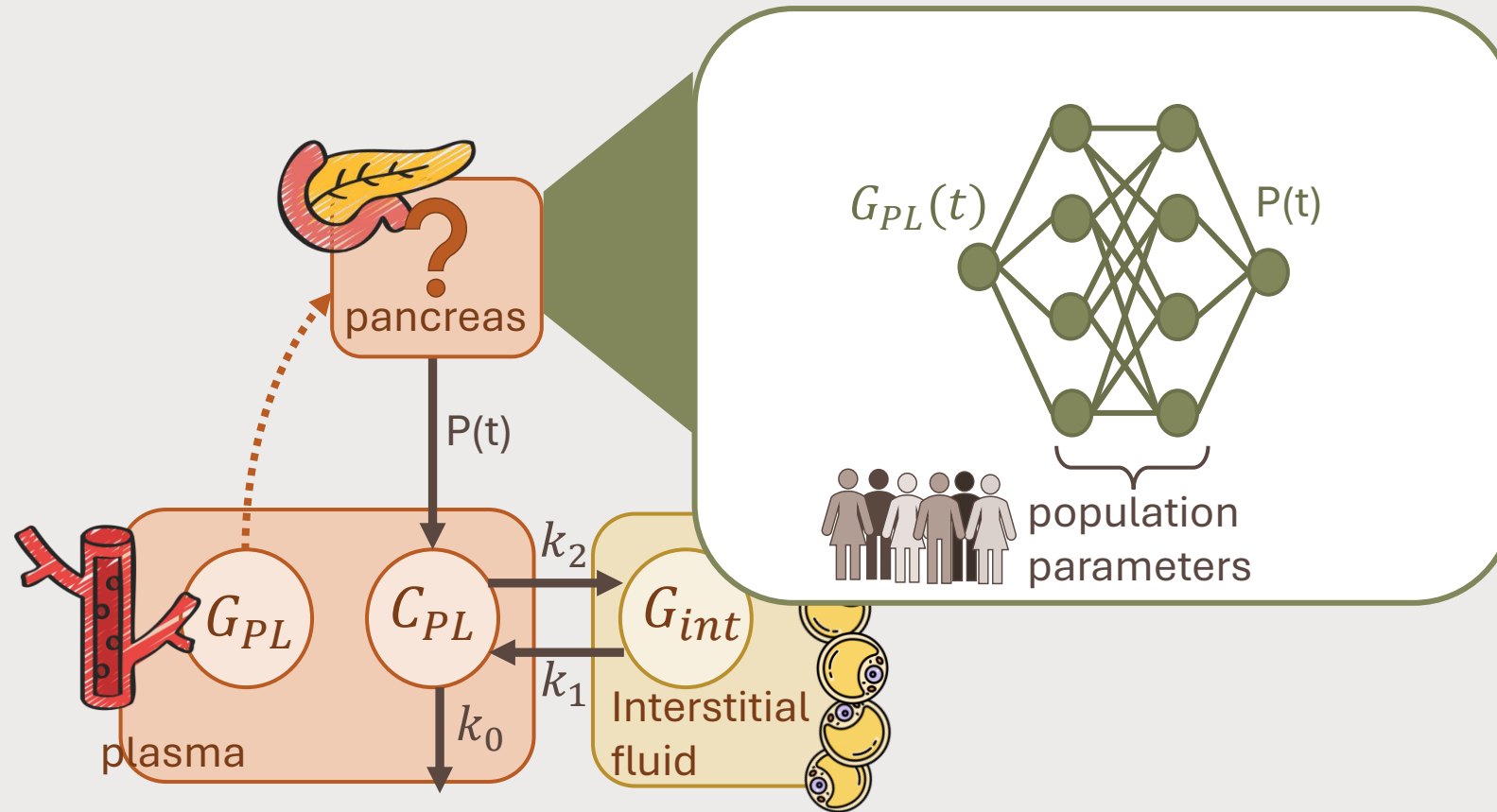
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Universal Differential Equations in Biology



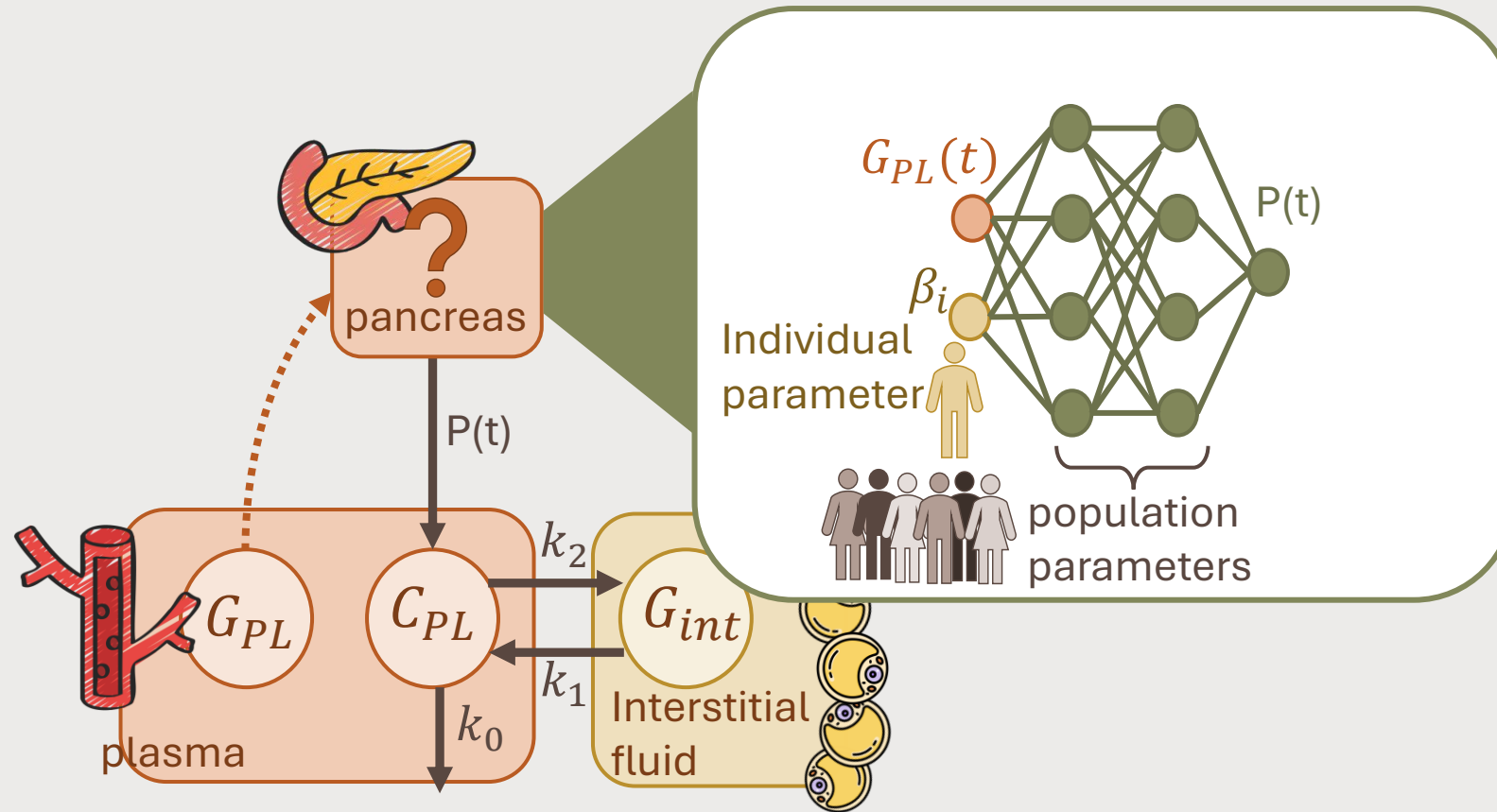
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Universal Differential Equations in Biology



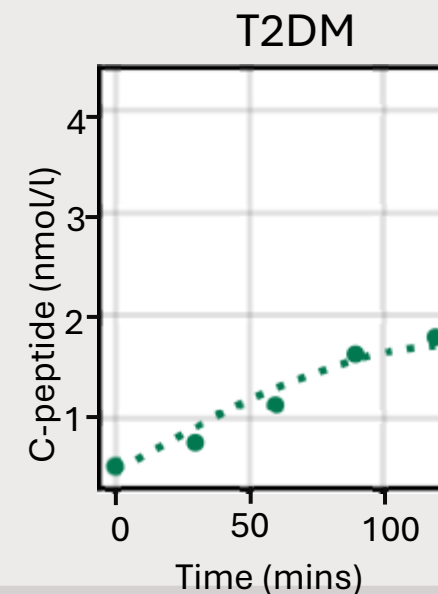
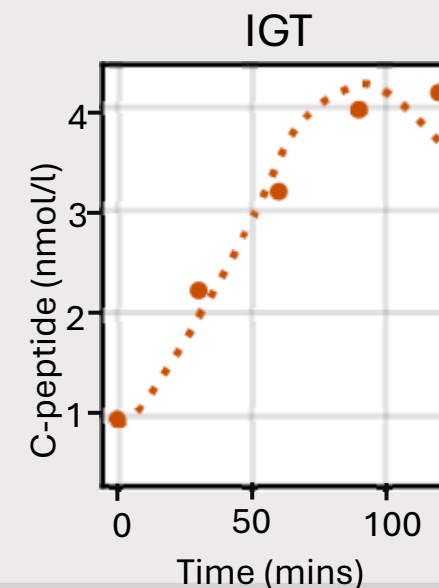
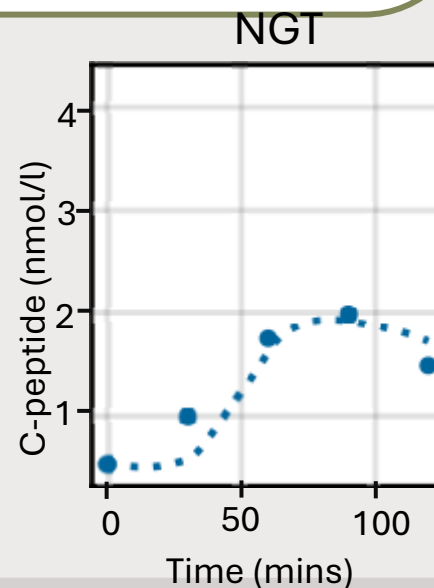
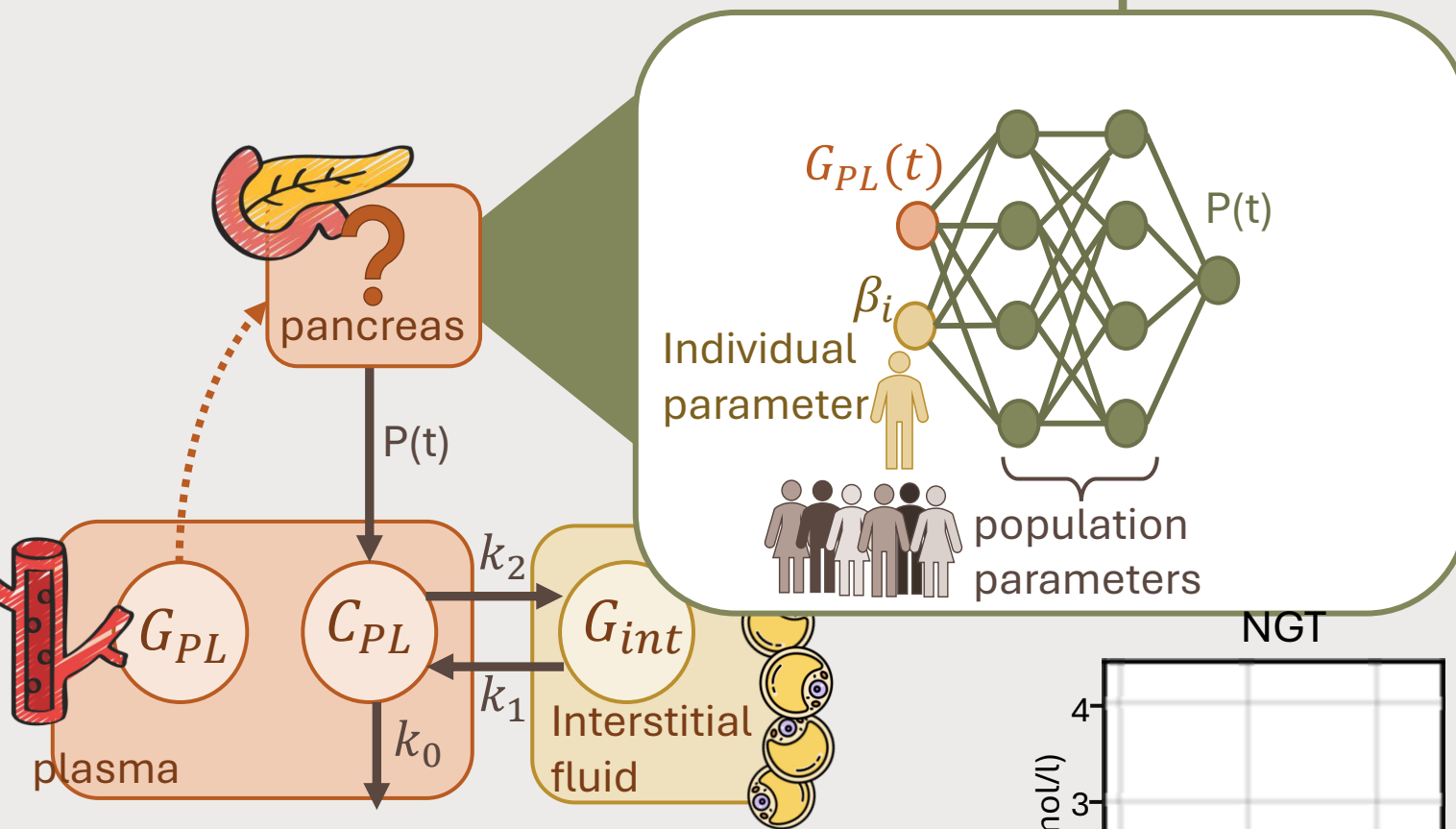
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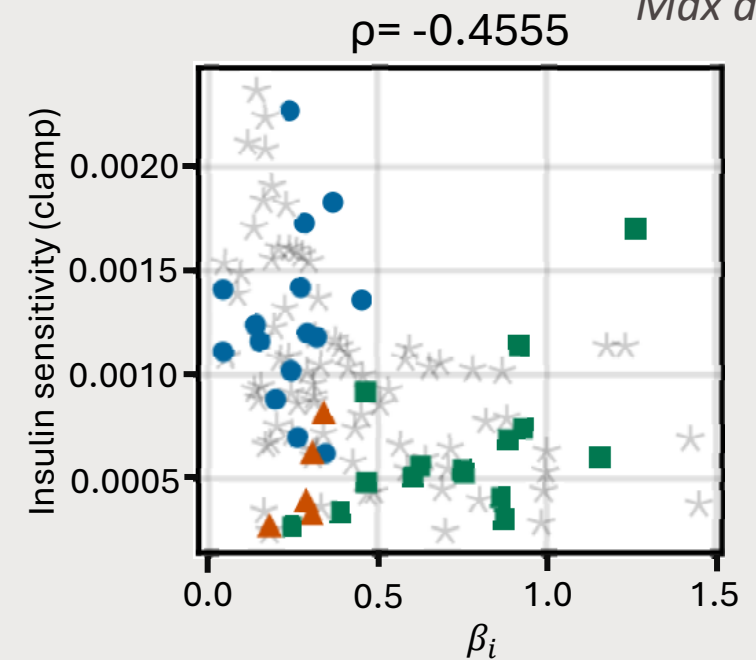
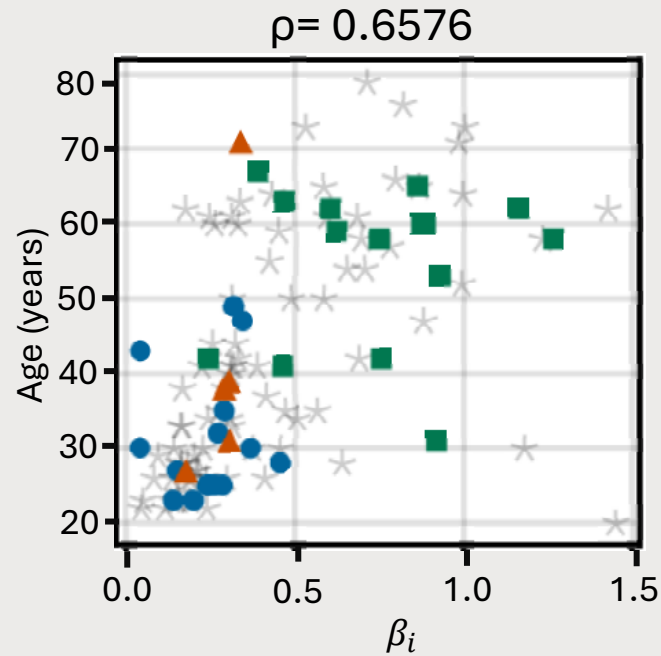
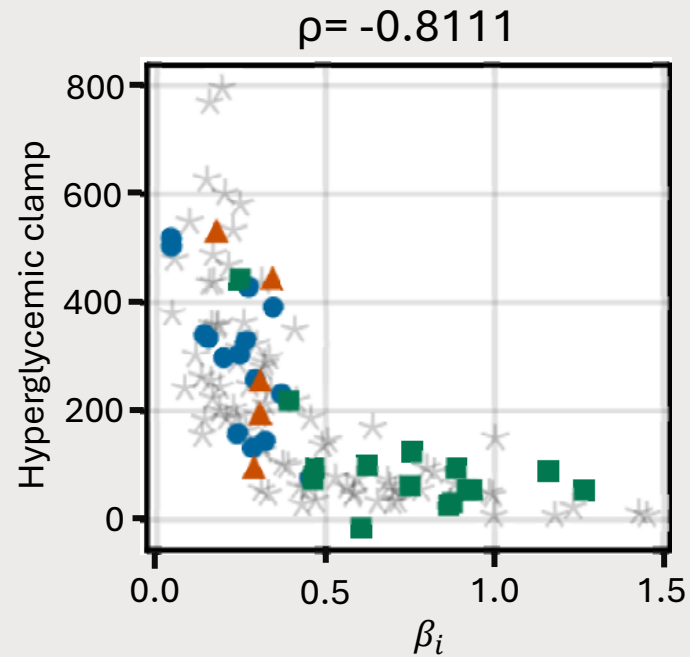
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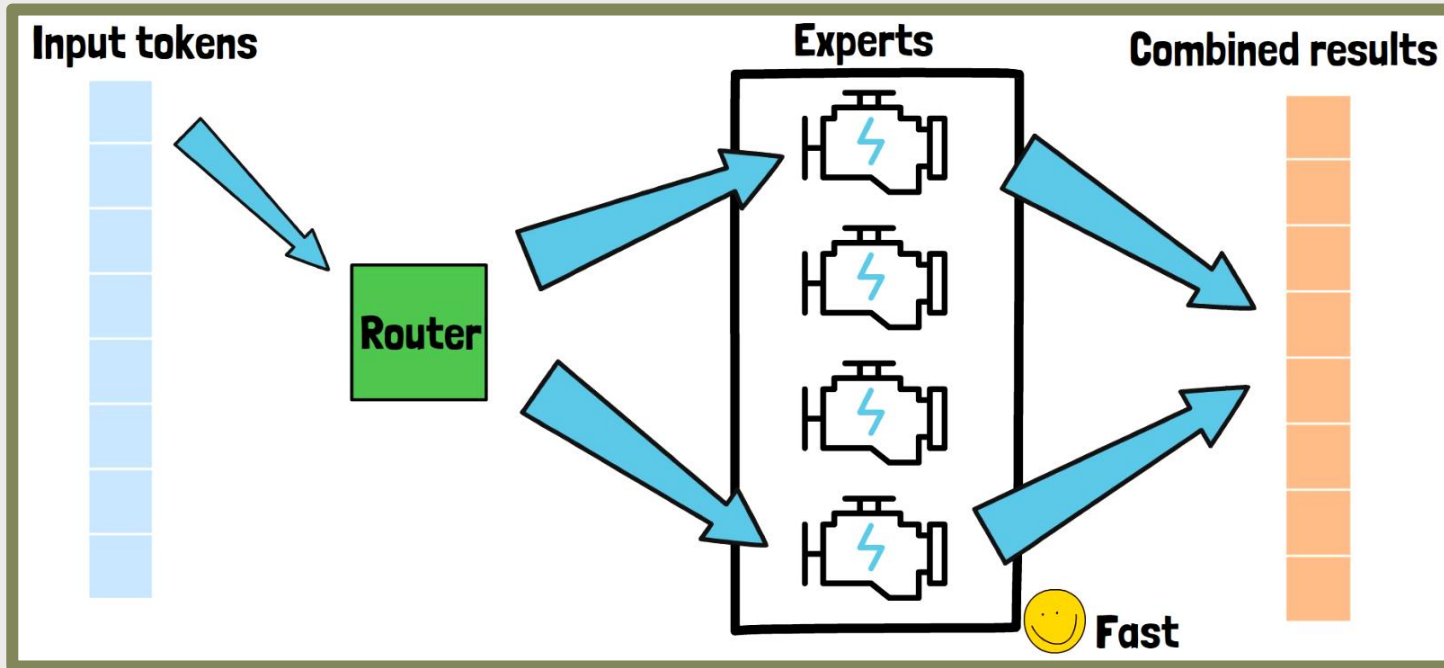
Universal Differential Equations in Biology



Max de Rooij



Mixture of Experts



Scope of problem

We want a method that can train a reliable personalised model for all people

Challenge

- Sparse data (infrequent sampling)
- Noisy data
- Heterogeneity
- Small sample size

